# Female Urology

*Condensed rapid revision · facts already answered in Papers 1–4 point to that answer*

**highlighted** number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book


| Chapter | Topic |
|---|---|
| Ch 101 | Surgical Anatomy of the Female Pelvis |
| Ch 108 | UI & Prolapse: Epidemiology & Pathophysiology |
| Ch 105 | Evaluation of Women With UI & POP |
| Ch 114 | Conservative Management of UI |
| Ch 116 | Retropubic Suspension (Burch) |
| Ch 118 | Slings: Autologous, Biologic, Synthetic, Mid-Urethral |
| Ch 117 | Reconstructive Surgery for Pelvic Organ Prolapse |
| Ch 119 | Mesh Complications & Their Repair |
| Ch 121 | Urinary Tract Fistulae |
| Ch 122 | Bladder & Female Urethral Diverticula |
| Ch 77 | Female Sexual Function & Dysfunction |
| ★ | Putting it together: female pelvic decisions |

# 101 Surgical, Radiographic and Endoscopic Anatomy of the Female Pelvis

## 1 · Perineum and external genitalia

- Urethral meatus **1 cm** anterior to the vaginal opening.
- **Skene's (paraurethral) glands**: beside the meatus; prostate homologue. Obstructed periurethral glands → diverticulum or cyst.
- **Bartholin's gland**: under posterior end of each vestibular bulb — cyst/abscess.
- **Deep perineal space** (above membrane): urethral sphincter, urethrovaginal sphincter, compressor urethrae, deep transverse perineal. **Superficial space** (under Colles fascia): superficial perineal muscles, clitoris, vestibular bulbs, Bartholin's.
- **Perineal body**: bulbospongiosus, external anal sphincter, superficial and deep transverse perineal converge. Episiotomy angled laterally to spare the sphincter.
- **Labia majora: dual supply** — superficial external pudendal (femoral) above, posterior labial (internal pudendal) below → reliable **Martius flap**.

## 2 · Bony pelvis and ligaments

- **Ischial spine** (≈ S5): palpate sacrum → SSL → spine; landmark for apical sutures.

| Fixation point | Used for | Adjacent danger |
|---|---|---|
| **Sacrospinous ligament (SSL)**: ischial spine → lower sacrum/coccyx | SSL fixation (apical prolapse) | Pudendal nerve/vessels near spine; sacral plexus just posterior |
| **Cooper's ligament** (medial superior pubic ramus) | Burch colposuspension for SUI | Obturator vessels; aberrant obturator |
| **Obturator foramen**; canal superolateral | Transobturator tape (TOT) trocar | Obturator nerve/vessels → groin pain, bleeding |
| **Anterior longitudinal ligament (ALL)** over S1 | Sacrocolpopexy mesh | Presacral veins, middle sacral vessels, left common iliac vein, right ureter |
| **Uterosacral ligament** | Uterosacral (apical) suspension | Ureter (lateral, closest at cervix); sacral nerve entrapment |

- Pudendal nerve leaves via **greater**, re-enters via **lesser sciatic foramen** to Alcock's canal.

## 3 · Pelvic floor muscles

- Pelvic diaphragm = **levator ani** + coccygeus; levator supplied directly by **S3–S5 (mainly S4)**.
- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *levator table: puborectalis, pubococcygeus, iliococcygeus; coccygeus NOT levator, lies on SSL*
- **Levator hiatus** transmits urethra, vagina, rectum; **levator plate** = horizontal midline shelf for viscera.
- Muscle/nerve injury → lax hiatus → plate tilts → **prolapse**.

## 4 · Endopelvic fascia and visceral ligaments

- The **intermediate** fascial stratum (parametrium, paracolpium, pubocervical fascia) **gives most organ support**.
- **Pubourethral ligaments**: pubis → bladder neck/urethra; puboprostatic homologue; with anterior vaginal wall form a mid-urethral hammock; weakness → SUI.
- Vesicopelvic fascia defect at arcus tendineus → paravaginal cystocele.
- **Cardinal (Mackenrodt's)**: carries internal iliac branches and pelvic plexus — division at hysterectomy → bladder dysfunction.
- **Infundibulopelvic** (ovarian vessels; ureter beneath) vs **ovarian ligament** (ovary → uterus) MCQ.

### DeLancey levels of vaginal support †

- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *I suspension (uterosacral–cardinal; uterine/vault prolapse, enterocele), II attachment (ATFP; anterior/posterior wall), III fusion (urethral hypermobility, distal rectocele, perineal descent)*

> **KEY POINT**
> Support ligaments are connective tissue condensed around vessels and nerves — the best anchors (uterosacral, cardinal) are exactly where sutures injure ureter, pelvic plexus and sacral nerves. → *Ch 108*

## 5 · Urethra and continence anatomy

- Length **4.7–5.1 cm**; behind symphysis; suspended by posterior pubourethral ligaments; rests on anterior vaginal wall.
- Submucosa has a **spongy vascular plexus**; epithelium transitional → squamous distally.
- **Mucosa/submucosa generate most closure pressure and are oestrogen-dependent** (rationale for vaginal oestrogen).
- **Internal sphincter**: smooth (detrusor continuation), autonomic, bladder neck/proximal urethra. **External**: striated, horseshoe at mid-urethra, voluntary (pudendal), distal two-thirds.
- Striated complex: sphincter urethrae, **compressor urethrae** (closes urethra against anterior vaginal wall), **urethrovaginal sphincter** (narrows hiatus).
- **Hypermobility** is one cause of SUI. Autologous pubovaginal sling at the **bladder neck**; synthetic mid-urethral sling (MUS) at the **mid-urethra**.

## 6 · Bladder and ureter

- **Dome** (peritoneum): intraperitoneal injury → formal repair. **Base** (anterior vagina, lower uterus) → vesicovaginal/vesicouterine fistula.
- **Pelvic ureter**: over iliac bifurcation → behind ovary → base of broad ligament → **under the uterine artery** ("water under the bridge") → over anterior fornix → bladder.

> **ALGORITHM**
> **Ureteric danger points**: pelvic brim (infundibulopelvic ligament) → under uterine artery (hysterectomy) → beside cardinal/uterosacral near cervix → anterior vaginal fornix (vaginal surgery) → sacral promontory (right ureter, sacrocolpopexy)

## 7 · Nerves at surgical risk

- **Pudendal nerve (S2–S4)** crosses SSL/coccygeus **close to the ischial spine** → Alcock's canal; branches inferior rectal, perineal, dorsal nerve of clitoris.
- **Pudendal block**: transvaginal, at SSL just behind the spine.
- SSL fixation: sutures ≈ 2 cm medial to the spine, through the ligament, not around it †.

| Nerve | Procedure / mechanism | Deficit |
|---|---|---|
| Common peroneal | Lithotomy: compression at fibular head | Foot drop |
| Sacral plexus (L4–S4) | SSL fixation (exits just posterior to SSL) | Posterior leg motor/sensory loss |
| Pudendal | SSL fixation near spine | Perineal pain, sensory loss, sphincter dysfunction |
| Sacral trunks S1–S4 | Uterosacral suspension (deep sutures) | Buttock/posterior thigh pain |
| Ilioinguinal (L1), genitofemoral (L1–L2) | Hernia repair, caesarean, hysterectomy | Mons/labial numbness or pain |
| Obturator (L2–L4) † | TOT trocar, obturator lymphadenectomy | Groin pain, adductor weakness |
| Posterior femoral cutaneous (S1–S3) | Gluteal pressure, pelvic tumour | Perineal sensory loss |

**KEY POINT**

Post-lithotomy foot drop is peripheral (fibular head), not pelvic. Leg pain and swelling after long lithotomy → **compartment syndrome**: emergency († fasciotomy).

## 8 · Vessels and lymphatics

| Internal iliac | Anterior division | Posterior division |
|---|---|---|
| Visceral | Obliterated umbilical, superior vesical, uterine, vaginal, middle rectal | None |
| Parietal | Obturator, internal pudendal, inferior gluteal | Iliolumbar, lateral sacral, superior gluteal |

- Obturator artery may arise from inferior epigastric.
- **Internal iliac**: rich collaterals → ligation/embolisation tolerated. **External iliac**: poor collaterals → ligation threatens the limb.
- Ovarian vein: right → IVC, left → left renal.
- **Presacral plexus**: valveless, high pressure (worse in lithotomy); a 1–2 mm vein can bleed > **1 L/min**.

**ALGORITHM**

**Sacrocolpopexy**: expose ALL at promontory over L5/S1 disc → open peritoneum distally → suture into ALL over **S1** (strongest; less discitis) → avoid left common iliac vein, middle sacral vessels, presacral plexus, right ureter (each ≈ 2.7 cm from mid-promontory)

| Organ | First nodes |
|---|---|
| Distal ⅓ urethra, lower ⅓ vagina, vulva | Inguinal → pelvic |
| Proximal ⅔ urethra, upper ⅔ vagina | Iliac, obturator (pelvic) |
| Bladder, uterus | Perivisceral → pelvic → presacral → great vessels |
| Ovary MCQ | Along gonadal vessels → para-aortic (bypasses pelvis) |

## 9 · Endoscopic anatomy

> **ALGORITHM**
> Distal urethra (glands, diverticular ostia, mesh, stones) → mid-urethral coaptation (external sphincter) → bladder neck coaptation → trigone and orifices (30° lens) → whole-mucosa survey

- Marked anterior/apical prolapse: reduce with a vaginal sponge stick.
- Squamous metaplasia, cystitis cystica: benign, but biopsy if doubt.

## 10 · Imaging

- **Translabial ultrasound**: bladder-neck mobility, prolapse; shows **mesh, MUS, bulking agents**; 3D adds axial view.
- **Endoanal US**: internal sphincter defect = bright break in dark muscle; external = dark lesion in bright muscle.
- **MRI**: most comprehensive radiation-free. **T2-weighted MRI is best for urethral diverticulum** and to separate it from other cysts (Gartner, Skene, Bartholin, müllerian, epidermal inclusion).
- **MR defecography**: descent graded below the **pubococcygeal line** (inferior symphysis → last coccygeal joint); for defaecatory dysfunction or symptoms out of proportion to examination.

# 108 Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology

## 1 · Definitions and terminology

- **Urinary incontinence (UI)**: complaint of involuntary loss of urine (ICS/IUGA, 2010).
- LUTS are **non-specific** → combine **signs**, **symptoms** and **observations** (diary, UDS).
- **Bother drives treatment**: no QoL impact → may need no treatment.

| Type | Complaint of loss… | Urodynamic correlate / trap |
|---|---|---|
| **Stress UI (SUI)** | On effort, cough, sneeze | UD SUI: leak with rise in abdominal pressure **and no detrusor contraction** |
| **Urgency UI (UUI)** | With urgency | **DO not required**; DO incontinence (DOI) is a separate UD diagnosis |
| **Mixed UI (MUI)** | With urgency and effort | Urge-, stress-predominant or equal |
| Overflow UI | Frequent small-volume loss with retention | Underactivity or obstruction; high PVR |
| Functional UI † | Impaired mobility/cognition, normal LUT | Frail elderly, long-term care |
| Continuous UI | Day and night, few voids | Suspect fistula |
| Nocturnal enuresis | During sleep | ≠ nocturnal UUI (wakes, cannot reach toilet) |

- **Urgency** = sudden compelling desire, hard to defer (pathological) vs **urge** = normal desire. **Increased sensation**: early desire, voiding can be postponed.
- **OAB**: urgency ± UUI, usually with frequency and nocturia; **defined by urgency**. Frequency ≥ **8 voids/24 h** is abnormal once polyuria excluded.

## 2 · Epidemiology of female UI

- Incidence (40–60 y) 1–10%/year; remission up to ≈12%/year, **lowest for MUI**.
- **Pregnancy**: commonest cause of incident UI in young women (≈ **40%**, mostly SUI). >30% leak at 1 year; UI at 3 months usually persists. PFMT during/after pregnancy reduces UI.
- **Hysterectomy** can cause or cure UI; vaginal/laparoscopic > open.

## 3 · Risk factors for female UI

| Factor | Key point |
|---|---|
| **Ageing** | Strongest independent non-modifiable factor. SUI peaks 40–59 y; UUI/MUI dominate >60. Long-term care: >40% at admission |
| **Vaginal delivery** | Causes later **SUI, not UUI**. CS protects only if **every** delivery is CS; NNT **13** to prevent one moderate–severe UI. Elective CS < CS after labour. Baby ≥4 kg: OR ≈1.5. |
| **Parity** | First delivery OR ≈1.5; modest rise thereafter; strongest for SUI |
| **Obesity** | BMI ≥30 more than doubles risk (SUI, MUI). Dose/duration-dependent, **reversible** with weight loss. Little effect on SUI surgery success |

| Factor | Key point |
|---|---|
| **Menopause / oestrogen** | SUI rises across the transition. **Systemic oestrogen increases UI** (incident UI nearly doubled vs placebo); vaginal oestrogen not linked to SUI, treats atrophy and recurrent UTI |
| Race | White > Black/Asian for SUI (higher closure pressure in Black women) |
| Smoking | Urgency, frequency, UUI (dose-dependent) |
| Caffeine | UUI, MUI, OAB — not SUI |
| Diabetes | Up to 2×; neuropathy, macroalbuminuria independent factors |

KEY POINT

Obstetric injury and loss of support → **stress** UI; ageing, neurological, metabolic and lifestyle factors → **urgency**. So dominant type moves from stress to urge with age.

### Male UI (contrast)

- **UUI > MUI > SUI**; male SUI almost always iatrogenic (RP, pelvic surgery) or neurological. Remission up to **40%**.
- Post-RP UI 8–60% by definition; better with bladder-neck, nerve and Retzius sparing and youth. TURP ≈1%.

## 4 · Physiology of continence

### Low-pressure storage

- Filling: **pelvic** parasympathetic suppressed; **hypogastric** sympathetic β → detrusor relaxation, α → bladder-neck closure; **pudendal** (Onuf's) tonic sphincter; suprapontine centres inhibit the pontine reflex until voiding is appropriate.
- **Poor electrical coupling** of detrusor cells stops local contractions spreading; ECM maintains compliance (altered by inflammation, radiation).

### Outlet

- Three components: **mucosa** (seal), **internal sphincter**, **distal rhabdosphincter**.
- **Female**: continence zone **proximal and mid-urethra**; bladder neck often incompetent in continent women. Oestrogen-sensitive spongy submucosa gives up to **30%** of closure.
- **Support**: anterior vaginal wall = firm backing; pubourethral ligaments limit proximal movement. Lax vaginal support with intact pubourethral ligaments → **shear** in the continence zone.
- **Male vs female**: male bladder neck powerful (α, prostate) and outlet fixed, so loss of one unit is tolerated (TURP); female bladder neck weak, outlet mobile — external sphincter deficiency → UI.

KEY POINT

The female outlet relies on a **single** mid-urethral sphincter against a mobile backing — continence needs support, intrinsic urethral properties and muscle together.

## 5 · General pathophysiology: bladder vs outlet

| Mechanism | Causes | Result |
|---|---|---|
| Loss of suprapontine inhibition | Stroke, MS, Parkinson; early DM | Neurogenic DO → UUI |

| Mechanism | Causes | Result |
|---|---|---|
| Outlet obstruction | Women: after anti-incontinence surgery; men: BOO | DO → de novo UUI |
| Detrusor underactivity | Conus/cauda, neuropathy (DM, alcohol), radical hysterectomy, APR, opioids incl. tramadol | Retention → **overflow UI** |
| Poor compliance | Pelvic radiotherapy | High detrusor LPP → UI |
| Sphincter deficiency | RP, urethral/anti-incontinence surgery, cord injury, α-blockers, muscle relaxants | ISD → SUI |

- **DM**: early polyuria and UUI → later loss of sensation/contractility → poor emptying, UTI, overflow.
- Multiple system atrophy: intrinsic sphincter destroyed; ≥20% UI after TURP — counsel.
- Post-RP UI: **ISD predominates (>70%)**; DO/compliance contribute in >30%.
- Each failed urethral/anti-incontinence operation adds fibrosis and ISD.
- **Labour**: fascial injury, ischaemia, **pudendal denervation**.

## 6 · Pathophysiology of female SUI

- P3 P1 Female pelvic floor, continence mechanism, classification and management of SUI — *theories table: pressure transmission (Enhorning), hammock (DeLancey; repositioning not essential), integral theory (Petros–Ulmsten → MUS), ISD (McGuire)*
- Urethral hanging theory: proximal urethra hangs below a less mobile bladder neck and pulls it open (funnelling). Integral theory also: premature stretch-receptor firing → urgency.

### Hypermobility vs ISD

- ISD described with the autologous pubovaginal sling, usually after previous surgery; = **type III SUI** (Blaivas) MCQ.
- **"Pipestem" urethra**: fixed, non-coapting, usually iatrogenic. Subtler ISD coexists with hypermobility in most SUI.
- **VLPP**: abdominal pressure causing leak (invalid during detrusor contraction or poor compliance). **VLPP < 60 cm $H_2O$ = ISD**; historically → bulking or pubovaginal sling.

> KEY POINT
> **Slings work in hypermobility but Burch fails in ISD**: almost all SUI includes some ISD, so sphincter-directed procedures (PVS, MUS, bulking) work in hypermobile women; Burch, correcting only hypermobility, is weak with severe ISD and a fixed urethra.

## 7 · OAB pathophysiology (brief)

- Neurogenic (§5) or outlet obstruction (BOO, post-sling) → DO and de novo urgency.
- † Non-neurogenic: **myogenic** (altered excitability and coupling), **urotheliogenic** (mediator release, afferent sensitisation), **afferent** (increased signalling).
- With POP: frequency tracks anterior/apical POP; UUI tracks anterior POP severity. Main mechanism: kinking BOO. **OAB improves after POP repair in 60–80%**, more with anterior/apical repair; no reliable predictor.

## 8 · Insensible incontinence

- Usually ISD or DOI; if treatment fails, seek rarer causes (UDS may miss them).

- Diverticulum: post-void dribble → pelvic MRI. Ectopic ureter: continuous loss with normal voids → upper-tract imaging. Overflow: frequent small loss → UDS/PVR.

## 9 · Pelvic organ prolapse: definition and staging

- **POP**: descent of uterus/vault and/or vaginal walls from loss of support. **Name the vaginal segment** (anterior, apical, posterior), not the presumed organ. Enterocele: peritoneal sac with bowel in the rectovaginal space.
- Most specific symptom: **visible/palpable bulge**; correlates best once past the hymen.
- Never attribute postmenopausal bleeding or pain to POP. **Pessary trial** is diagnostic: relief predicts surgical benefit.

| POP-Q stage | Most distal point on maximal strain |
|---|---|
| 0 | Aa, Ba, Ap, Bp = −3; C/D ≤ −(tvl − 2) |
| 1 | > 1 cm above hymen (< −1) |
| 2 | Within 1 cm of hymen (−1 to +1) |
| 3 | > +1 but < +(tvl − 2) |
| 4 | Complete eversion: ≥ +(tvl − 2) |

- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *POP-Q points: hymen 0; Aa/Ap 3 cm (−3 to +3), Ba/Bp, C, D, gh, pb, tvl; 3 × 3 grid*
- Aa ≈ urethrovesical junction. Traps: stage = single most distal point; Ba/Bp = C in total eversion; measuring right after pessary removal underestimates.
- Baden–Walker grades 0–4: poor interobserver reliability; POP-Q is the IUGA/ICS-recommended system.

## 10 · POP epidemiology and risk factors

- Loss of support in 43–90%; **asymptomatic stage 1–2 is normal**. Symptomatic POP 3–12%.
- **Anterior > posterior > apical**, but apical descent is under-diagnosed — accompanies most significant anterior POP, universal at stage 4. **Planning cystocele repair → assess apex, consider vault suspension.**
- Slow progression, early regression common. Progression predictors: age >60, Ba ≥4 cm, gh ≥5 cm.

| Risk factor | POP | UI (contrast) |
|---|---|---|
| **Vaginal childbirth** | Strongest; rises steeply with parity; forceps worse. Elective CS ≈ nulliparous; CS after labour ≈ vaginal | SUI, not UUI |
| Age | ≈40% rise per decade; surgery peaks 7th decade | Shift to UUI/MUI |
| Obesity | ≈1.4–2.5-fold | >2-fold, reversible |
| Hysterectomy | Raises risk, most if done for POP (omitted vault suspension) | Cause or cure |
| Race | Hispanic highest, Black lowest | White highest |
| Oestrogen / menopause | Not consistently associated | Oral oestrogen ↑ UI |
| Other | Family history (≈5× in siblings), connective-tissue disease, chronic constipation (2.5×) | — |

- Recurrence after surgery predicted by preoperative stage 3–4.

## 11 · Pathophysiology of POP

- **Levator ani** is the primary support; uterosacral–cardinal ligaments hold organs in position over it.
- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *DeLancey I suspension (uterine/vault, enterocele), II attachment to ATFP (lateral or central anterior defect), III fusion (urethral hypermobility, distal rectocele)*
- ATFP: pubis (1 cm from midline) → just above ischial spine. Commonest cystocele defect at surgery: ATFP detachment from the ischial spine.
- Defects rarely isolated; level I loss correlates strongly with anterior POP — apex is the keystone.

> **ALGORITHM**
> **POP sequence**: levator trauma/denervation (labour) → loss of tone → widened hiatus → fascia/ligaments carry whole load → stretch and fail (ageing, straining, hysterectomy) → POP → hiatus widens further

- Levator defects on MRI in up to **20%** of primiparas (none in nulliparas); **avulsion** → advanced, multicompartment POP.
- Levator EMG neuropathy in ≈ a quarter after vaginal delivery or CS in labour; almost none after elective CS — **labour itself is the injury**.
- Collagen type I:III ratio altered; Ehlers–Danlos, Marfan increase risk.

## 12 · POP and UI interaction

- 40–55% of SUI have significant anterior POP; ≈20% of OAB.
- **Occult SUI**: seen only with prolapse reduced (up to a quarter of advanced POP).

> **ALGORITHM**
> Cystocele → urethral kink → obstruction masks SUI → POP repair relieves kink → SUI unmasked. Detect by UDS / stress test **with prolapse reduced** (pessary or speculum).

- De novo SUI after POP surgery: age >66, DM, low urethral closure pressure.
- Missed POP at sling surgery → later prolapse surgery: examine for POP in every UI evaluation.

## 13 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| CARE | Stress-continent women, abdominal sacrocolpopexy ± Burch | Burch reduced postoperative SUI; SUI on reduction UDS predicted it | Prophylactic Burch at sacrocolpopexy prevents de novo SUI |
| OPUS | Stress-continent women, vaginal POP repair ± prophylactic MUS vs sham | UI at 1 year 27% vs 43% (NNT ≈6); more bladder perforation, UTI, bleeding, retention | Fewer de novo SUI, more adverse events — counsel |

# 105 Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse

## 1 · Definitions and risk factors

- **UI** risk factors: age, obesity, smoking, comorbidity, postmenopausal hormone therapy; vaginal delivery → SUI.
- **POP**: **anterior > posterior > apex**. Half of parous women have anatomical POP; disease = **symptoms attributable to it**. **Vaginal bulge**: most specific symptom, most likely to resolve after repair.

| Category | POP risk factors |
|---|---|
| Primary only | Older age, high BMI, high birth weight, levator defects, low education, poverty |
| Recurrence only | **Younger age**, preoperative POP-Q stage 3–4 |
| Both | Vaginal delivery, parity, levator hiatal ballooning |
| Protective (primary) | Caesarean delivery; **smoking** MCQ |

- **ICS UI terms** (by trigger): SUI, UUI, MUI; **nocturnal enuresis** = involuntary *voiding* in sleep; **disability-associated** — cannot reach toilet; **insensible** — unaware; **occult SUI** — only after POP reduction; also coital, continuous, postural.

## 2 · History

- Ask — women rarely volunteer. Exclude reversible causes — **DIAPPERS**: Delirium, Infection, Atrophic vaginitis, Psychological, Pharmacological, Excess urine output, Restricted mobility, Stool impaction.
- **SUI = urethral failure; UUI = bladder failure; MUI = both**. In MUI treat the more **bothersome** component first. Negative SUI history does not exclude SUI.
- LUTS may precede a neurological diagnosis.
- **POP**: bulge, pressure, worse later in day; symptoms start when it reaches the hymen. **Splinting** → high-grade POP; anterior POP kinks urethra → voiding difficulty; posterior → obstructed defecation. POP-associated OAB may persist after repair.
- SUI coexists in ~**40%** with POP; patients often expect repair to cure leakage → counsel.
- Past history: neurological disease, DM, radiation, pelvic surgery, prior UI/POP surgery or mesh, obstetric trauma; **local** oestrogen may improve UI, **systemic** may worsen. Family history raises POP risk and recurrence.

**Occult SUI** (ICS-defined; ~⅓ before vaginal POP repair) vs **de novo SUI** (after POP surgery, none occult preoperatively; not ICS-defined) → shared decision on concurrent vs staged surgery.

- **Drugs**: α-agonists → retention; **α-blockers → worsen SUI in women**; anticholinergics, calcium-channel blockers, opioids → poor emptying; cholinesterase inhibitors → UUI; diuretics, SGLT2 inhibitors, lithium → polyuria; glitazones → nocturia; sedatives → confusion; **ACE inhibitors → cough → SUI**.

## 3 · Physical examination

- Frailty assessment preoperatively at ≥65 y. **Oestrogen status**: atrophic = pale, flat, dry, no rugae. Note caruncle, urethral mass.

### SUI testing

- AUA/SUFU 2023: **objectively demonstrate SUI with a comfortably full bladder before surgery**.
- P3 SUI: evaluation in females, role of UDS, and surgical therapy in males — *cough stress test (supine, then standing), reduce prolapse for occult SUI, Q-tip >30° = hypermobility*
- CST at half functional capacity (largest diary void): high sensitivity and specificity; at comfortably full bladder lower specificity.
- Q-tip now mostly teaching/trials; vaginal swab test equivalent; **visual urethral mobility** (meatal rotation < or **>30°**) is routine.

### POP examination

- Lithotomy **and standing**, straining, for the worst prolapse.
- **Split Graves speculum**: retract posterior wall → anterior; rotate → posterior; both blades → apex. Blade expelled with posterior foreshortening = **level I loss → vault prolapse**; blade stays = isolated rectocele/enterocele.

### POP-Q system

- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *POP-Q points table: hymen 0; Aa/Ap 3 cm (−3 to +3), Ba/Bp, C, D (absent after hysterectomy), gh, pb, tvl*
- Ba/Bp range −3 to +tvl. gh = mid meatus → posterior hymen; pb = posterior gh → mid anus; tvl measured with prolapse reduced. Stage = single most distal point.

| Stage | POP-Q criteria | Baden–Walker grade (by eye) |
|---|---|---|
| 0 | Aa, Ba, Ap, Bp −3; C or D ≤ −(tvl − 2) | 0 — normal |
| I | Leading edge < −1 cm | 1 — upper vagina to midplane |
| II | Leading edge −1 to +1 cm | 2 — midplane to hymen |
| III | > +1 cm but < +(tvl − 2) | 3 — through/beyond hymen |
| IV | ≥ +(tvl − 2) — complete eversion | 4 — complete eversion |

> **KEY POINT**
> Stage II = 2-cm window around the hymen. III/IV boundary depends on tvl. Example: tvl 8, Bp +5 → between +1 and +6 → **stage III**.

### Other elements

- **Bulbocavernosus reflex** (S2–S4) present in only ~**70%** of normal women → absence alone not pathological.
- **Prior mesh** (IUGA/ICS): *prominence* (no breach) vs *separation* (breach); *extrusion* (through vaginal epithelium) vs *perforation* (into bladder/urethra/viscus).

## 4 · Supplemental evaluation

**Universal: UA and PVR.** AUA/SUFU 2023 index patient for SUI surgery: **UA, PVR and objective SUI demonstration** required; cystoscopy, UDS, imaging not routine. Non-neurogenic OAB: only UA required (PVR optional).

- P3 SUI: evaluation in females, role of UDS, and surgical therapy in males — *non-index features needing more tests: SUI not shown, neurogenic, urgency-predominant MUI, raised PVR, stage ≥3 POP without SUI, failed surgery; questionnaires; VALUE*
- Also: unexplained haematuria/pyuria.

### Symptom quantification

- **Bladder diary** (3 days): diagnostic and therapeutic → *Ch 114*. Largest void = functional capacity.
- ICIQ-UI SF correlates with pad tests; **PFDI / PFIQ** for POP.
- **Pad tests** (mainly research). ICS 1-hour: 500 mL in <15 min, then standard activities → weigh.

| | 1-hour test | 24-hour test |
|---|---|---|
| Normal | <1.4 g | <4.4 g |
| Mild | 1–10 g | 4–20 g |
| Moderate | 11–50 g | 21–74 g |
| Severe | >50 g | ≥75 g |

- Pad *counts* do not correlate with 24-hour pad weight.
- **Double-dye tampon test** (oral phenazopyridine + intravesical methylene blue): orange = ureterovaginal, blue = vesicovaginal fistula.
- UA: haematuria, pyuria, glucosuria; don't treat asymptomatic bacteriuria.
- **PVR**: bladder scan preferred; repeat (single value unreliable). No agreed cut-off; asymptomatic women almost always <100 mL. Raised PVR alters counselling before SUI surgery or antimuscarinics.
- **UDS**: not needed in the index SUI patient → *Ch 107*.
- **Cystoscopy** not routine; for haematuria, urgency/irritative LUTS, recurrent UTI, prior sling/mesh, pelvic radiation.

### Imaging

| Modality | Main indication |
|---|---|
| Voiding cystourethrogram | Reflux, diverticula, cystocele |
| Renal ultrasound | Hydronephrosis: chronic retention, **severe POP (ureteric kinking)**, fistula, neurogenic bladder |
| Translabial ultrasound | **Modality of choice for mesh** (CT/MRI poor); levator defects |
| MRI / dynamic MRI | **Urethral diverticulum (standard of care)**; complex/recurrent POP, symptoms out of proportion |

- Diverticulum **3 Ds**: dysuria, postvoid dribbling, dyspareunia → *Ch 122*. MRI **HMO**: H-line = hiatal width; M-line = levator plate descent; O = organ beyond H-line.

## 5 · Management pathway

**ALGORITHM**

History + examination + UA + PVR → exclude DIAPPERS, UTI, obstruction, tumour → classify, grade bother → red flags (pain, haematuria, recurrent UTI, pelvic mass, prior radiation/pelvic surgery, prolapse beyond introitus, fistula) → specialist work-up; otherwise initial management → reassess → extra tests only if they change management → surgery.

| Diagnosis | Initial management |
|---|---|
| SUI | Lifestyle, behavioural therapy incl. PFMT → *Ch 114*; † pessary; surgery if these fail |

| Diagnosis | Initial management |
|---|---|
| UUI / OAB | Lifestyle, behavioural; antimuscarinic trial → *Ch 113* |
| Significant PVR | Treat constipation, review drugs; catheter drainage if PVR 200–500 mL, then reassess |
| Symptomatic POP | † Observation, PFMT, pessary, or surgery by bother and goals |

## 6 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| † VALUE | RCT, uncomplicated demonstrable SUI: office evaluation ± UDS | Office evaluation alone non-inferior | Preoperative UDS not needed in index SUI |

# 114 Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices

## 1 · Definition and principles

- **Conservative treatment** = any therapy that is neither surgery nor drugs.
- **Behavioural treatment** changes behaviour/environment or teaches skills; effective and **essentially risk free** → first-line in all guidelines for SUI, UUI, MUI, OAB and nocturia.
- Multicomponent, individualised: education, diary, PFMT, "the knack", urge suppression, bladder training (BT), biofeedback (BF), E-stim, scheduled voiding, fluid/caffeine/weight. Delivered mostly by nurses/APPs.
- Outcome unpredictable → offer to anyone willing; dementia patients benefit from caregiver-led programmes.

| | Outlet-directed | Bladder-directed |
|---|---|---|
| Target | Urethral closure | Bladder function via voiding habits |
| Examples | PFMT; behavioural training with urge suppression | BT; delayed voiding |

| | Patient-dependent | Caregiver-dependent |
|---|---|---|
| Examples | BT, PFMT, behavioural training | Timed voiding, habit training, prompted voiding |
| Needs | Intact function, learning, motivation | Knowledgeable, motivated caregiver |
| Suits | Cognitively/functionally intact | Impaired, nursing home |

**KEY POINT**

Why first-line if surgery cures SUI more often? Conservative therapy is risk free and at least as effective as drugs — this asymmetry of harm drives every recommendation.

## 2 · Assessment

- History targets *triggers and habits* (onset, progression, triggers) rather than diagnosis.
- **Bladder diary, 3–7 days**: intake, voids/volumes, urgency, leaks, pads. Estimates **functional capacity**; **self-monitoring effect** → it is assessment, outcome measure and treatment at once.
- **Questionnaire-Based Voiding Diary**: validated recall alternative, 5–7 min at first visit.
- **Education** first: anatomy, mechanism of own leakage; improvement gradual, needs daily practice.

## 3 · Lifestyle modifications

Usually adjunctive; evidence weaker than skill-based therapy, except weight loss.

### Fluid management

- **Inadequate intake is more common** than excess (self-restriction → dehydration, UTI, constipation, concentrated irritant urine, reduced capacity). Excess (~3.7 L/day) worsens frequency → reduce if 24-h output >~ **2100 mL**.

- **Target**: six to eight 8-oz glasses, or **30 mL/kg/day**, minimum **1500 mL**, at set times (IOM 2.7 L/day includes food water).
- **Timing**, not restriction: stop fluids **3–4 h before bed** for nocturia. Oedema-driven nocturia → stockings/leg elevation late afternoon; loop diuretic mid-to-late afternoon.
- Reducing intake by ~¼ cuts frequency, urgency, nocturia; goal is a **normal range, not restriction**.

### Caffeine and irritants

- Caffeine: **diuretic and irritant**; raises filling detrusor pressure, risk factor for DO. ~**250 mg/day** (≈2 cups) associated with moderate–severe UI in men; coffee not linked to nocturia.
- Only a subset is caffeine-sensitive: 3–5-day reduction trial; taper with decaf to avoid withdrawal headache.
- Other suspects (aspartame, citrus, spicy, carbonated/artificially sweetened drinks): eliminate one at a time with a food diary.

### Bowel function

- Constipation (<3 stools/week) and straining linked to LUTS (shared innervation, cross-sensitisation, straining → **pudendal denervation**). **Faecal impaction** → OAB or overflow.
- Fibre **with adequate fluid**; exercise; regular post-meal evacuation (gastrocolic reflex).

### Obesity, smoking, diet

- Obesity ~doubles UI risk. **Loss of 5–10% body weight** sustained 12 months substantially reduces UI (mainly SUI); bariatric surgery resolves UI in morbid obesity.
- Behavioural weight loss in overweight women = the **only grade A lifestyle recommendation**.
- Smoking: no UI recommendation possible (grade D); cessation for general health (grade C in men). Fruit/vegetable/whole-grain diet helps prevent UI postmenopausally (B); vitamin D reduced UUI in postmenopausal Black women (B).

## 4 · Pelvic floor muscle training (PFMT)

- Cornerstone for SUI, UUI, MUI, urgency, frequency. **Arnold Kegel (1948)** MCQ.
- **Mechanism**: contraction lifts and compresses urethra against symphysis; training → striated **hypertrophy** → higher urethral pressure; better bladder-neck support. Plus a **skill** route (the knack).

### Assessment of PFM function

- Correct contraction = **perineal elevation**; incorrect = perineal descent (straining).
- **Digital vaginal exam**: levator ani just above the hymen at **4 and 8 o'clock**; tone, symmetry, hold, quick flicks. Scales: **Modified Oxford** (most used); **Brink** (pressure, duration, displacement). Anal: puborectalis 2.5–4 cm from verge.
- Note co-contraction and **overactivity** (hypertonic, cannot relax) vs hypotonicity.

### Teaching

- "Tighten as if stopping gas, pull up and in" without buttocks/abdomen; men — "move the penis up and down".
- Commonest errors: **co-contraction** (rectus, gluteals) and **Valsalva substitution**; count aloud to avoid breath-holding.

- **Verify correct contraction before prescribing** — failure to isolate is probably the commonest cause of poor outcome; a pamphlet or "stop the stream" does not ensure it.

### Regimen

No ideal regimen; good results with **45–60** contraction–relaxation pairs/day.

> **ALGORITHM**
> Empty bladder → **lying**: 5 quick (1–2 s) + 5 sustained (5 → 10 s) → **sitting** → **standing** (= 30) → rest as long as each contraction (work:rest **1:1 to 1:2**) → twice daily = **60/day**.

- Maximal effort per contraction; progress reclining → upright.
- **The knack**: contract **just before and during** cough/sneeze/lift; immediately reduces leakage. If leaking, squeeze anyway.

### Evidence

- PFMT > no treatment for SUI, UUI, MUI, short and long term; pure SUI best.
- **Supervised > unsupervised** (grade A); **age no deterrent**.
- Pregnancy: antenatal PFMT in continent women prevents late-pregnancy/postpartum UI; supervised postpartum PFMT if UI persists ~3 months.
- **Post-RP**: PFMT hastens continence in first 3 months; 8 weeks behavioural training helped men leaking 1–17 years after RP. Adding E-stim/BF does not improve continence. Pilates/yoga not substitutes.
- Post-micturition dribble: strong PFM contraction after voiding or urethral milking.
- **Neurological OAB** (MS, incomplete SCI, Parkinson, stroke): PFMT safe, first-line (B); TENS first-line option after stroke. PFMT adds to tolterodine for UUI.

### Cones and devices

- **Weighted vaginal cones**: slipping prompts reflex contraction (progressive overload); 2 × 20 min/day. **Cannot be retained with POP or wide introitus**; some retain without contracting. Not better than PFMT; adding them adds nothing.
- Colpexin sphere: no added strength. **Motion-sensing inserts** distinguish lift from strain (pressure perineometers cannot); app-guided device beat written/video programme at 12 months.

## 5 · Urge suppression

- A well-timed PFM contraction **reflexly inhibits the detrusor**. Rushing raises bladder pressure; toilet cues are conditioned triggers.

> **ALGORITHM**
> Urge → do not rush → stand still/sit, relax → several quick PFM squeezes → distract → urge subsides (a wave) → walk normally to void → once mastered, wait 5 min, then extend.

- Stand-alone for UUI: mean **60–80%** reduction in incontinence; also urgency, frequency, nocturia. **Nocturia**: suppress on waking; void if not passed in 1–2 min.
- ≥ antimuscarinics; adds to drugs. Men: lowered frequency more than antimuscarinic + α-blocker; on α-blocker, adding behavioural therapy reduced nocturia more than adding antimuscarinic.

## 6 · Biofeedback (BF)

- A **teaching technique**, not a treatment: real-time display of PFM activity (manometry or surface EMG); operant conditioning. Kegel's perineometer = first BF device. EMG shows PFM vs abdominals; pressure BF cannot separate lift from strain.
- EMG: fast-twitch ~2-s spikes; slow-twitch ~10-s plateaus.
- **Adds no clear benefit to supervised PFMT** (older benefit = extra contact). Use for very weak PFMs, poor proprioception, uncertain contraction, motivation.

## 7 · Electrical stimulation (E-stim)

- Non-implanted (vaginal/anal probe, surface): **PFM contraction** and **detrusor inhibition**.

| | Long-term (chronic) | Maximal |
|---|---|---|
| Intensity | Below sensory threshold | Just below pain threshold |
| Aim | Detrusor inhibition via pudendal afferents | Urethral closure (direct and reflex striated contraction) |
| Dose | 6–12 h/day for months | 15–30 min, several times a week to twice daily |

| Indication | Frequency |
|---|---|
| UUI | **5–20 Hz** (low) |
| SUI (incl. post-prostatectomy) | **20–50 Hz** (higher) |
| MUI | ~20 Hz or alternating |
| All | Pulse 200–1000 µs, rectangular, biphasic |

- > sham but **adds nothing to PFMT**; PFMT first-line (A). **Niche**: women who **cannot contract voluntarily**.
- **Tibial nerve stimulation** for UUI/OAB: transcutaneous = percutaneous efficacy, fewer adverse effects → *Ch 115*.

**KEY POINT**
E-stim dosed by goal: sub-sensory, low-frequency, prolonged → detrusor inhibition; near-painful, higher-frequency, brief → sphincter contraction. Volitional PFMT does both, so E-stim earns its place only where volitional contraction is absent.

## 8 · Bladder training and scheduled voiding

- P1 Describe pathophysiology of OAB. Briefly outline management & describe nonmedical non-pharmacological management — *BT breaks urgency → frequent voiding → reduced capacity cycle; void by clock not urge (dissociates voiding from urgency)*
- Components: education, **incremental schedule**, urge control (distraction, "I can wait", quick flicks).

**ALGORITHM**
Diary → initial interval ~1 h awake (30 min if needed) → void on waking, at each interval and bedtime, not between → increase **15–30 min/week** if leaks and urgency controlled → target **3–4 h** → weekly review with reinforcement.

- > no treatment for UUI, SUI, MUI; first-line in women (A); insufficient evidence in men. AHRQ: BT = PFMT. BT with prompting after hip fracture reduced functional UI at 6 months.

- **Delayed voiding**: no schedule; at each urge suppress, **wait 5 min**, extend gradually (BT triggered by clock; DV by urge + wait).

### Caregiver-dependent toileting — elderly, nursing homes

| Programme | Patient role | Key features |
|---|---|---|
| Timed voiding | Passive | Fixed 2–4 h by day; dryness goal. Also impaired sensation, infrequent voiders. ~20% of frail elders dry, 30–50% improve; 2–3-day trial predicts |
| Habit training | Passive | Diary-individualised, shorter than usual interval, **pre-empts** expected leaks; diuretic or time-patterned UI |
| Prompted voiding (PV) | Responsive | **Monitoring → prompting → praise**; every 2–2.5 h. 25–40% of nursing-home residents respond (daytime leaks ≤1); 3-day trial predicts |
| Functional incidental training | Responsive | PV + mobility/strength exercise |
| Bladder training | Active | Lengthens interval; needs ability, motivation |

- **Only BT lengthens the interval**; caregiver programmes aim at dryness. PV needs a consistent caregiver; success depends on caregiver more than patient. Ultrasound-assisted PV beat fixed-interval PV.
- † First exclude reversible causes (DIAPPERS) and check PVR → *Ch 105*.

## 9 · High-tone PFM dysfunction

- Raised tone, poor relaxation → pelvic/bladder pain, hesitancy, straining, incomplete emptying. Valsalva voiding provokes reflex PFM contraction; normal voiding starts with PFM relaxation.
- **Down-training**: contract then relax; work:rest **1:4 or longer**; BF useful. Voiding: slow down, breathe, relax, wait; double voiding.

## 10 · Mechanical devices

**Intravaginal** (support) or **intraurethral/meatal** (occlude); no device proven superior.

### Continence pessaries

- **Rationale**: restores proximal urethra above the floor, **stabilises bladder neck, lengthens urethra, compresses it against the pubis** — like a sling. Ring knob sits at the urethrovesical junction.
- **Types** (SUI): incontinence **ring** (knob), **dish**, **Hodge**. † POP: support (ring, Gehrung) and space-filling (Gellhorn, cube, donut) for advanced POP or poor perineal support.
- † **Fitting**: largest comfortable, one finger-breadth from wall; walk, cough, strain, void before leaving — no expulsion, pain or obstruction.
- Continuous or exercise-only; also helps POP-related urgency/frequency.
- **Contraindications**: active pelvic infection, severe ulceration, silicone/latex allergy, **likely non-compliance with follow-up**.
- **Complications**: discharge, odour; neglected use → **erosion, impaction, vesicovaginal/rectovaginal fistula**, bleeding.
- **Care**: follow-up every **24 weeks** acceptable (vs 12; grade C); anaesthetic cream eases removal; oestrogen + Lactobacillus adds nothing. † Inspect for erosion each review.

- ~Half improve; ring pessary for OAB ≈ behavioural therapy. ICI: pessary or PFMT may be offered; pessary + PFMT > PFMT for POP symptoms. **ATLAS** — trials.

### Other devices

- **Uresta**: reusable bell-shaped pessary with handle (sizes 3–5), wide base supports urethra. **Impressa**: disposable tampon-like, self-inserted; tension-free suburethral support; need not be removed to void.
- Urethral inserts/meatal devices: none available in USA. † Insert = balloon plug for pure SUI (UTI, haematuria, migration).
- † Pads when treatment fails/declined; prevent **incontinence-associated dermatitis**. Female external catheters for bed-bound.
- † **Indwelling catheter is the last resort**: unmanageable retention, urine-contaminated sacral/perineal wounds, terminal care, preference. CISC preferred for retention.

## 11 · Vaginal oestrogen

- SUI with atrophy: PFMT + vaginal oestrogen > oestrogen alone (C). † Local oestrogen improves urgency, frequency, recurrent UTI and pessary tolerance → *Ch 113*.

## 12 · Adherence, prevention and delivery

- **Adherence caps efficacy**; declines over time. Remedies: realistic expectations, follow-up, reminders, linking to daily cues. Predictors: self-efficacy, intention, therapist enthusiasm.
- **Prevention**: older women — group classes; **childbearing** — supervised antenatal PFMT (12 weeks from 16–24 weeks' gestation) prevents late-pregnancy and early postpartum UI (A); postnatal programme after forceps or baby ≥**4000 g**; antenatal PFMT does **not** prevent postnatal prolapse; **prostatectomy** — pre ± early post-op PFMT shortens time to continence.
- **Delivery** limited by access, not efficacy: continence APPs, group classes, internet programmes (maintained at 2 years), evidence-based apps (Tät); most apps poor. Intensity traded for reach.

## 13 · Evidence grading — key ICI recommendations

| Grade (Oxford) | Meaning |
|---|---|
| A | Consistent level 1 evidence; effectively mandatory |
| B | Consistent level 2/3 studies or majority of RCT evidence |
| C | Level 4, or Delphi-processed expert opinion |
| D | **No recommendation possible** (not "weak") |

| ICI 2023 recommendation | Grade |
|---|---|
| Supervised PFMT first-line for UI in women of all ages | A |
| PFMT over E-stim as first-line | A |
| BT first-line for women with UI | A |
| Behavioural weight loss for overweight women with UI | A |
| Antenatal supervised PFMT in continent women to prevent UI | A |
| PFMT reduces POP symptoms (stage reduction uncertain) | A; B |
| PFMT over drugs first-line (fewer adverse effects, less dropout) | B |

| ICI 2023 recommendation | Grade |
|---|---|
| PFMT before surgery (surgery more effective but riskier) | B |
| Adding BF, cones or E-stim to PFMT — not routine | B/C |
| Fluid adequacy and caffeine reduction | B |
| Smoking cessation for UI | D |

- AHRQ (2018): behavioural therapy **more likely than anticholinergics to cure UUI** (high strength).

## 14 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| First behavioural vs drug RCT | Older women, UUI: behavioural vs titrated drug | Behavioural fewer leaks, more satisfaction | Behavioural ≥ drugs for UUI |
| MOTIVE | Men with OAB, no obstruction: behavioural vs oxybutynin ER | Equivalent reduction in voids | Behavioural therapy works in men |
| UITN (†BE-DRI) | Urgency-predominant UI: tolterodine ER ± behavioural | Combination better satisfaction, bother; persisted after drug stopped | Add behavioural training to drugs |
| ATLAS | Stress-predominant UI: pessary vs behavioural vs both | 3 mo: behavioural better satisfaction; combination not better; no difference at 12 mo | Pessary reasonable alternative; behavioural first |
| Weight-loss RCT (†PRIDE) | Overweight women: 6-month intensive weight loss vs education | Greater UI reduction | Recommend 5–10% weight loss (A) |
| GBT | Women with UI: one 2-h bladder-health class vs none | Modestly better ICIQ-SF to 12 months | Group classes cheap, scalable |

# 116 Retropubic Suspension Surgery for Incontinence in Females

## 1 · Mechanism

- SUI mechanisms: **urethral hypermobility** and **ISD**, alone or combined. Suspension rests on **pressure transmission** (Enhorning); hammock (DeLancey) = compression against a firm backstop.

> **KEY POINT**
> Suspension corrects **hypermobility, not the sphincter**: it lifts and stabilises bladder neck and proximal urethra, so presupposes a mobile urethra. A fixed ISD urethra needs compression (sling, AUS).

## 2 · Patient selection

- Stress test supine → standing if negative; Q-tip >30° = hypermobile; POP, oestrogen status, prior incisions.
- AUA/SUFU minimum: **PVR + urinalysis**; UDS only if leak not shown. VLPP <60 or MUCP <20 cm $H_2O$ suggests ISD. For Burch, **mobility matters more than urethral function**.
- **Indications** (Burch remains an AUA/SUFU option with MUS, autologous sling, bulking): SUI with hypermobility or stress component of MUI (not pure UUI); **concomitant abdominal surgery** (sacrocolpopexy); **difficult vaginal access**; **mesh aversion**.

| Contraindications | Raise failure risk |
|---|---|
| Immobile vagina (cannot reach Cooper's) | Baseline urgency, preoperative anticholinergic |
| Immobile urethra (negative Q-tip ≈ **5-fold** failure) → sling or AUS | Obesity, menopause, age >60 |
| ISD alone; "lead-pipe" urethra | Prior hysterectomy or SUI surgery; advanced POP; MUI |

## 3 · Anatomy

- **Retzius**: avascular preperitoneal space, symphysis–bladder; floor = endopelvic fascia. **Cooper's ligament** on superior pubic ramus. **ATFP**: symphysis → ischial spine; lateral vaginal detachment = **paravaginal defect**.

## 4 · Burch colposuspension

Paravaginal tissue sutured to Cooper's; sutures elevate, **fibrosis maintains**.

> **ALGORITHM**
> Low lithotomy, prep incl. vagina, Foley → Pfannenstiel/low midline (± Cherney) → enter Retzius, sweep fat laterally → bladder neck by Foley balloon + vaginal finger → dissect bladder off vagina **laterally** → expose Cooper's → sutures → cystoscopy

- P1 Burch colposuspension — *2–4 sutures (0/1-0) each side, full thickness not lumen; distal at mid-urethra, proximal at bladder neck, lateral; complication rates*
- Sutures medial → lateral; tie **distal first**, **loosely** (Tanagho) — **≈ 2 fingerbreadths** between bladder neck and pubis; over-tightening → obstruction.

KEY POINT

**Danger points**: never dissect midline over urethra/bladder neck (Santorini bleeding, injury, obstruction if sutures medial). Passing through Cooper's may bleed suddenly — pressure, clips, sutures.

| Approach | Bottom line |
|---|---|
| Open (Pfannenstiel) | Cure **85–90% at 1 y, ≈70% at 5 y** |
| Laparoscopic (1991) / robotic | Subjective cure equivalent to open to 18 months |
| Needle suspension (Pereyra, Stamey, Raz) | Abandoned — sutures tore through |

## 5 · Other retropubic procedures

| Technique | Anchor | Holds by | Obstruction |
|---|---|---|---|
| Burch | Cooper's ligament | Sutures; high V-shaped angle | Greater |
| MMK | Symphyseal periosteum | Sutures; high | Greater |
| Paravaginal (Richardson) | ATFP | Scarring; flatter | Less |
| VOS | Obturator fascia (± iliopectineal) | Scarring; flatter | Less |

- MMK unique complication: **osteitis pubis**.

KEY POINT

AUS in women: second-line for ISD (recurrent/neurogenic). At bladder neck, preferably **abdominally**; transvaginal raises infection and erosion.

## 6 · Perioperative care

- **Cystoscopy mandatory**: no suture in bladder/urethra; bilateral efflux — if absent, release sutures.
- Incomplete emptying: CIC (preferred to suprapubic) until PVR consistently **<150 mL**.

## 7 · Outcomes and complications

- Burch ≈ retropubic MUS; meta-analyses slightly favour MUS (more perforation); Burch longer, slower recovery, more voiding dysfunction. PVS more effective, more obstructive (SISTEr). Long-term failure ≈ 1 in 5 both.
- With POP: prophylactic Burch at sacrocolpopexy in stress-continent women (CARE); with existing SUI, MUS beats Burch.
- Voiding dysfunction 2–27% → CIC, urethrolysis → *Ch 118*; de novo OAB 8–27% → exclude obstruction; **posterior prolapse** 2.5–27% (anterior elevation opens cul-de-sac) → examine posterior compartment.
- **Osteitis pubis** (MMK 0.7–2.5%): 2–12 weeks, suprapubic pain to thighs → rest/analgesia; osteomyelitis → antibiotics, debridement.
- Follow-up by 6 months (continence, voiding, OAB, sexual function, PVR); long-term = >48 months.

## 8 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| SISTEr | RCT, 655: Burch vs autologous fascial PVS | PVS higher cure at 2 and 5 y; more UTI, voiding dysfunction, de novo UUI | Sling more effective, more morbid |
| Ward–Hilton (TVT vs colposuspension) | First RCT, 348 | No difference to 5 y; TVT more perforation; Burch more voiding dysfunction, later POP | MUS equals Burch, less invasive |
| CARE | RCT, 322 stress-continent, sacrocolpopexy ± Burch | Less postoperative SUI with Burch, sustained 2 y | Add prophylactic Burch at sacrocolpopexy |

# 118 Slings: Autologous, Biologic, Synthetic and Mid-Urethral

## 1 · How slings work — theories of continence

- P3 Female pelvic floor, continence mechanism, classification and management of SUI — *hammock (DeLancey) vs integral theory (Petros–Ulmsten: pubourethral ligaments, suburethral hammock, pubococcygeus); PVS at bladder neck, MUS loose at mid-urethra*
- Hammock: deficient suburethral layer (attached to ATFP) → hypermobility → unequal pressure transmission. Integral: maximal urethral closure pressure (MUCP) at mid-urethra; injury (parturition, surgery, ageing, oestrogen loss) to any component → SUI.
- Integral theory support: incontinent women have ≈ **40% less collagen** in skin/round ligament biopsies MCQ.
- **Urethral hypermobility is a SIGN** of damaged support, **not the cause** of SUI.
- Pubovaginal sling (PVS): **minimal tension at the bladder neck** rebuilds the hammock. Mid-urethral sling (MUS): **loose under mid-urethra**, arrests the posterior wall, aids coaptation. Neither obstructs at rest.

### Historical milestones MCQ

- Fascial PVS reintroduced 1978 (McGuire); short free rectus graft, adjustable tension (Blaivas).
- Intravaginal slingplasty 1995 → **TVT**, first retropubic MUS; first series defined the two complications **unique to MUS**: urinary tract mesh penetration and vaginal mesh exposure.
- Transobturator (Delorme, 2001) — avoids bladder, bowel, vessel injury. SIMS first approved 2006 (TVT-Secur, withdrawn).

### Guideline status (AUA/SUFU)

- MUS for index patient — **strong recommendation**, grade A. Retropubic or transobturator — either (conditional, grade A). SIMS — conditional, grade B.
- Discuss mesh-specific risks **and** non-mesh alternatives before synthetic MUS (Clinical Principle).

## 2 · Preoperative assessment and patient selection

- P3 SUI: evaluation in females, role of UDS, and surgical therapy in males — *AUA/SUFU minimum evaluation (history, exam, cough stress test supine then standing, PVR, urinalysis); indications for UDS; ValUE; ISD cut-offs*
- **Baseline urgency** predicts poorer sling outcome. Bother and goals drive the decision (QoL, not survival).
- Q-tip hypermobility = **>30°** on straining.
- **Cystoscopy** NOT part of index evaluation; do for haematuria, suspected neoplasm/foreign body, prior pelvic surgery. Upper-tract imaging/MRI only if indicated.
- Further evaluation (incl. UDS) when: unclear diagnosis, neurogenic, urgency-predominant MUI, high PVR, POP-Q ≥3 without SUI on reduction, prior surgery.

### Urodynamic studies (UDS) — the controversy

- May be **omitted** in index patient with demonstrable SUI (conditional, grade B) — **ValUE**. Criticism: most screened women ineligible; fellowship-trained surgeons.

- MUCP **<20 cm $H_2O$** = type 3 SUI = **ISD**; ALPP/VLPP **<60 cm $H_2O$** = ISD; **>90** = little/no ISD.
- ISD is a **matter of degree** (all SUI has some); now largely clinical. Low MUCP/VLPP predicts failure inconsistently (mainly transobturator); matters most with a **hypomobile urethra**.
- **DO** is the most consequential bladder finding — may improve (stress-induced) or worsen; consider treating first. Low capacity/compliance worsen outcomes. Detrusor underactivity/Valsalva voiding predict voiding difficulty but should **not exclude** surgery.
- Non-surgical options for all: weight loss (dose–response), **PFMT** before surgery for most (adherence limits), pessaries, bulking → *Ch 120*.

## 3 · Choosing the sling

| Situation | Preferred option | Reasoning |
|---|---|---|
| Index SUI, mobile urethra | MUS (RP or TO); SIMS acceptable | Equal efficacy, least morbidity |
| ISD with mobile urethra | Retropubic > transobturator | Meta-analyses favour retropubic |
| Fixed, immobile urethra | Retropubic MUS, autologous PVS, bulking, adjustable sling | AUA Expert Opinion; mobility predicts MUS success |
| Recurrent SUI after failed MUS | Retropubic MUS or autologous PVS | Retropubic better than TO for recurrence |
| Mesh complication, urethral injury, fistula, diverticulum, reconstruction | **Autologous PVS** | No mesh near damaged urethra; adds tissue |
| Declines mesh | Autologous PVS or Burch → *Ch 116* | — |
| Neurogenic SUI on CIC | Autologous PVS (may be tight) | Only once UDS confirm safe capacity/compliance |
| Concomitant POP surgery | MUS, PVS or Burch | Conditional, grade C |

## 4 · Pubovaginal sling

- **Indications**: primary/recurrent SUI from **hypermobility and ISD**; stress-predominant MUI; anteroapical POP; neurogenic SUI; **urethral reconstruction** (trauma, mesh perforation, hypospadias, fistula/diverticulum interposition); failed Burch or MUS.
- **Autologous fascial PVS = gold standard for all forms of SUI.**
- Gives **dynamic compression without obstruction**; too much tension → ischaemia, retention, erosion. Give-way prevented by **incorporation into endopelvic fascia with fibrosis**.
- Width **<2 cm** — compresses without a constricting band.

| Material | Behaviour | Status |
|---|---|---|
| **Autologous** — rectus, fascia lata | Best ingrowth/neovascularisation, minimal reaction; negligible erosion | **Gold standard**; cost: harvest pain, wound complications |
| Allograft | Processing weakens; heterogeneous, less durable | Mostly withdrawn |
| Xenograft (porcine, SIS, bovine) | Weaker, encapsulation; cure falls with time; porcine inferior in RCTs | Mostly withdrawn |

| Material | Behaviour | Status |
|---|---|---|
| Synthetic at bladder neck | Urethral perforation **15×**, vaginal exposure **14×** (esp. woven polyester) | **No longer used at bladder neck** |
| Vaginal wall sling | Weak; inclusion cysts, shortening | Historical |

KEY POINT

**Fibrosis paradox**: more fibrosis should mean better support, but the inflammation driving it also degrades the sling → erosion. Autologous fascia remodels with minimal inflammation — the ideal balance.

- **Rectus vs fascia lata**: equal strength and cure. Fascia lata: no abdominal wound problems, less blood loss; but repositioning, thigh pain. Prior **ventral hernia repair** → fascia lata. Narrow (1.5 cm) strips → no thigh herniation; Crawford stripper reduces pain.

## 5 · Autologous PVS — operative procedure

- **Counselling**: voiding dysfunction, de novo urgency; teach CIC in selected patients; won't fix bowel symptoms.
- Single-dose 1st/2nd-generation cephalosporin (cefazolin); IPC ± heparin. Lithotomy nerve injury (lateral femoral cutaneous, femoral, peroneal) rare, almost all resolve.

ALGORITHM

Pfannenstiel 6–7 cm → mark **2 × 8 cm** rectus graft → excise → close fascia (No. 1 PDS) → clean graft → No. 1 PDS stitched to each end → saline

- Transverse incision **≥2 cm above the symphysis** for tension-free closure; if impossible (rare) bridge with mesh/biologic.
- Fascia lata: incision above patella over iliotibial band, stripper proximally; graft 8 cm (classically 20 × 2 cm); **fascia lata not closed**; compression wrap 8 h.
- Vaginal: **inverted-U incision**, thick flaps; **empty bladder**, perforate **endopelvic fascia** toward ipsilateral shoulder; finger-dissect retropubic space. Reoperation: sharp dissection hugging pubic periosteum.

ALGORITHM

Bladder empty → Stamey needles **from above** on back of pubis into vaginal incision → **cystoscopy (70° lens)** → injury: repass **more laterally**, re-scope each pass → confirm ureteric efflux → graft centred → ends tagged above

- Do POP repair **before tensioning**; **remove weighted speculum**.
- **Tension**: tie over rectus with **two fingerbreadths** slack — deliberately **loose**; tighter only if lifelong CIC planned.
- Anchor graft to periurethral tissue (two 4-0 polyglactin); close vagina watertight; optional pack ± oestrogen cream.
- Postop: liberal fluids; pack and catheter out, trial of void day 1; failure → Foley/CIC, retrial within 5 days. No lifting >10 lb or intercourse for **6 weeks**.

## 6 · PVS outcomes

- Durable, even in complex patients (radiation, SCI, trauma); cure high even with MUCP <4 cm $H_2O$ → **severe ISD not a contraindication**.

- **Recurrent SUI**: repeat PVS or PVS after failed MUS (with partial mesh excision) succeed in most.
- Commonest "failure" = **storage symptoms/UUI**, not stress leak. De novo UUI 2–22%. Prior POP surgery predicted de novo urgency; tight technique, age, BMI, radiation did not.
- Four-quadrant modification: high cure; all failures from urgency/storage symptoms.
- Fascia lata vs rectus: equal at 1 y; rectus more wound complications, fascia lata less blood loss, transient thigh pain/numbness. Short (4 × 2 cm) grafts work in ISD.
- **Cochrane PVS vs MUS**: PVS probably no better, maybe less effective; MUS shorter, fewer complications (except bladder perforation), less voiding/storage dysfunction.
- Damaged urethra (after mesh excision/diverticulectomy) has poorest prognosis, but PVS still dries most.
- **MUI**: urgency outcome unpredictable; anticholinergics alone resolve urgency in only ≈ two-thirds. PVS cure in MUI ≈ pure SUI. PVS + anterior repair works for grade 3–4 cystocele with SUI.

## 7 · PVS complications and management

### Voiding dysfunction / BOO

- 2.5–35% — **more than Burch** (SISTEr: all outlet procedures in PVS arm). Retention >4 weeks ≈8%; **permanent ≤5%** (≈2% if not overtensioned).
- Cause: **excess tension**; host: poor contractility, dysfunctional voiding, **straining to void** (tightens the dynamic outlet).
- Presents as weak stream, retention, or **storage symptoms only** (urgency commoner than retention), recurrent UTI, pain — even with low PVR. Exclude new POP kinking the urethra.
- Preoperative prediction unreliable (PVR >100 mL, Qmax ≤20 mL/s inconsistent) → low detrusor pressure is reason to **counsel about CIC**, not refuse.
- Evaluation: exam (angulation, fixed urethra), PVR (no cut-off), cystoscopy, video-UDS selectively — high-pressure/low-flow often **absent in women**.

**KEY POINT**

**Key diagnostic criterion is the temporal relationship** between surgery and symptoms, not a urodynamic number. Preoperative UDS does not predict obstruction (SISTEr).

**ALGORITHM**

Transient retention (most void within **10 days**) → catheter/CIC, timed voiding, PFMT, anticholinergic → autologous obstruction often improves: wait up to **3 months** → complete retention, earlier: **<6 weeks** loosen autologous sling under anaesthesia (downward urethral traction) → **>6 weeks** or failed care → sling incision or urethrolysis

- Downward loosening **not for synthetic slings**. Dilation risks fibrosis. **Transurethral bladder-neck incision fails** (sling extraluminal) and risks sphincter damage.

| Procedure | Features |
|---|---|
| **Sling incision** (transvaginal) | 84–100% success; least morbid → **first choice**; ≈20% recurrent SUI; interposition doesn't prevent it |
| Transvaginal urethrolysis | Least effective for PVS — can't relieve suburethral compression |
| Suprameatal urethrolysis | Better access to lateral wings |
| Retropubic urethrolysis | Best results; after failed transvaginal attempt |

- Urethrolysis success 65–93%; recurrent SUI 0–19%; prior urethrolysis only suggested adverse factor. Failure usually **inadequate lysis**; repeat relieves obstruction in ≈90%.
- **Refractory storage symptoms** in >half: antimuscarinic/β3-agonist, **SNM** (test response predicts benefit), onabotulinumtoxinA 100 U (≈40% dry at 12 weeks).
- Counsel: lysis may not cure urgency and may make a dry patient leak; untreated female BOO has unknown long-term bladder effects.
- Recurrent SUI after lysis: bulking, repeat PVS or MUS.

### Perforation, exposure, other

- Rare with fascia (missed needle injury, excess tension, instrumentation). Urethral → **excise sling + repair** (± Martius); bladder → endoscopic removal. Bladder-neck sling perforations present 1–18 months (mean ≈9).
- Synthetic PVS exposure → removal; recurrent SUI 44–100% → second **autologous** PVS.
- Mortality ≈ **3 per 10,000** anti-incontinence procedures. Harvest-site **incisional hernia** ≈1%.

## 8 · Mid-urethral sling — mechanism and materials

- Mechanism: **secures the posterior urethral wall** ± compression and dynamic **kinking**. Anchorage: retropubic → endopelvic fascia; TO → obturator muscles/membrane. Fixed along **whole length** — why 50–93% stay dry after midline incision.
- Preoperative **immobility predicts failure**. Postoperative mobility not harmful (ideal = loose sling + mobile urethra). **Exception, SIMS**: success needs restricted mobility → tighter.
- Whether MUS obstructs is disputed; most surgeons tension retropubic and TO similarly, rarely with objective test.

> KEY POINT
> **Why does a loose tape work?** At rest it doesn't touch the urethra, so voiding is unobstructed. On coughing the mobile urethra descends onto the tape, the posterior wall is arrested, the urethra kinks and closes. A fixed urethra never meets the tape; tightening to compensate obstructs.

- Material: **type 1 (Amid) macroporous monofilament polypropylene**, pore **>75 µm** (macrophage entry, fibrovascular ingrowth). UraTape and ObTape (<50 µm) → high exposure → withdrawn.

## 9 · MUS operative procedures

- P2 P1 Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT; TOT in SUI — *TVT bottom-up steps, TOT outside-in/inside-out steps, tensioning, Cochrane complication rates*
- **Counsel**: permanent implant, mesh-specific complications, voiding/storage dysfunction, non-mesh alternatives; document.
- Cough-test tensioning under local unnecessary — same outcomes under spinal. Single antibiotic dose still given.

**ALGORITHM**

Local infiltration → two suprapubic stabs ≈2 cm from midline → 1.5 cm midline vaginal incision 1.5 cm from meatus → minimal lateral tunnels to endopelvic fascia (**not perforated**, unlike PVS) → catheter guide deflects bladder to opposite side → trocar through Retzius and rectus, **hugging the back of the pubis** → repeat → cystoscopy → tension loosely (clamp between tape and urethra) → remove sheath → trim

- TVT kit: two 5 mm trocars, 40 × 1.1 cm polypropylene tape. Local: ≈5 mL vaginal and suprapubic sites, 20 mL behind pubis, 10 mL each side of urethra.
- Keep trocar in **metal-to-bone contact** with back of pubis (extravesical, extraperitoneal). Bottom-up beats top-down: better subjective cure, less voiding dysfunction, perforation, exposure.
- TOT (outside-in): hips **hyperflexed (120°)**; dissect laterally to ischiopubic ramus; puncture labiocrural fold at adductor longus notch; trocar pierces obturator membrane ("give") → around ramus onto finger; check for vaginal sulcus button-hole.
- TVT-O (inside-out): exits 2 cm above urethral level, 2 cm lateral to labial folds; winged guide at 45°, helical passer. Same cure; inside-out → more voiding dysfunction, less vaginal perforation; outside-in possibly more urinary tract injury.
- Brief thigh flexion (≥70°) helps trocar passage.
- SIMS: vaginal incision only, anchored in **obturator internus**; tension **tighter** (hemostat tip passes); place close to urethra, wide deep tunnels.

| Feature | Retropubic (TVT) | Transobturator | SIMS |
|---|---|---|---|
| Path / anchor | Retzius, rectus fascia; U-shape | Obturator foramen; flatter hammock | Obturator internus only |
| Cure (index) | Equal to TO to 5 y; best long-term data (>17 y) | Equal; slightly more repeat surgery | Non-inferior to TO (TVT-Secur excluded) |
| ISD / recurrent | **Preferred** | Worse with low VLPP | Limited data |
| Bladder perforation | **4.5%** | **0.6%** | Rare |
| Vascular/visceral, bleeding, voiding dysfunction | More | Less | Less |
| Pain | Suprapubic 2.9% | **Groin 6.4%** vs 1.3% | Least |
| Vaginal exposure | ≈2% | ≈2% | Similar (excl. TVT-Secur) |
| Cystoscopy | Mandatory (70°) | Recommended; check urethra | Essential (30° suffices) |

## 10 · Intraoperative cystoscopy

- P1 Management of perforated mid urethral sling — *70° lens, full bladder; bladder injury: repass laterally and continue; urethral injury: no mesh, staged repair*
- Inspect the urethra too (TO, SIMS); bladder injury benign — catheter overnight to 72 h.

## 11 · MUS outcomes

- Retropubic: most studied (17–20 y); objective cure stays high, subjective lower; <85% on strict definitions. Cochrane (81 trials): RP = TO cure short to long term.

- TVT vs colposuspension: equal at 5 y; de novo urgency <2% vs <5%; later POP commoner after Burch.
- TOMUS 5 y: satisfaction high (79% RP, 85% TO) despite falling objective success.
- Long term: **de novo urgency rises** (≈⅓ at 10–20 y); retention 0–7%. A one-step fall in leakage frequency satisfies most.
- SIMS (Cochrane): worse than RP, driven by **TVT-Secur (withdrawn)**; others ≈ TO, less pain. Learning curve: failure 20% → 8% sited closer to urethra. Key long-term endpoint = **reoperation**.

| Group | Key points |
|---|---|
| **MUI** | Urgency resolves ≈60–75% (best stress-predominant). De novo urgency risk: age, obesity, nocturia, DO, low capacity. Persistent DO: TO < RP < PVS. New storage LUTS may be BOO |
| **ISD** | ≈50% cure with urodynamic ISD; immobile urethra stronger predictor than low VLPP; low VLPP not a contraindication to RP. TO: 12× odds of persistent SUI with low VLPP |
| **Recurrent SUI** | No level 1 evidence; repeat MUS commonest; RP better than TO; cure ≈81–90% but below primary (≈62% vs 86%); more urgency, more bladder perforation after prior RP |
| **Concomitant POP** | Concomitant sling ↓ postop SUI and reoperation but ↑ voiding dysfunction; ≈30% SUI resolves after POP repair alone; prophylactic sling only after counselling (OPUS); benefit only in RP subgroup in one review |
| **Elderly** | Safe; similar/lower cure, more urgency and voiding dysfunction. Q-tip >30° → 90% continent; <30° → 42%; <10° → 80% stay incontinent. Cure ≈64% at ≥75 y vs 89% at 55–64 y. **Frailty, not age**, predicts readmission. Don't exclude by age (AUA Expert Opinion) |
| **Obese** | Success >80% but lower (≈87% vs 66% at 5 y); risk factor for TVT failure; worse BMI >35; more wound infection; fewer bladder punctures |

## 12 · MUS complications and management

KEY POINT

**Concurrent vs staged sling at POP repair**. Concurrent: one operation, less SUI (≈39% → 8–19%), but more obstruction, storage symptoms, mesh events; tensioning hard if SUI occult. Staged: avoids unnecessary sling but commits a third to a second operation (usually within 12 months). Decide after testing for occult SUI with prolapse reduced.

- Bladder trocar injury ≈2–4%, almost all RP (↑ prior retropubic surgery, thin). Vascular (≈0.07%) and bowel (≈0.04%) injury rare but fatal cases reported.
- †Unexplained hypotension, falling Hb, pain, ileus or sepsis after RP sling → urgent imaging/exploration.
- Bleeding 2–3%, more RP (≈4% bled ≥300 mL needing intervention in one series); retropubic haematoma >5 cm in ≈1 in 8 on ultrasound, usually **asymptomatic → observe**.
- **UTI = commonest complication** (3–4% at 30 days; ≈1 in 4 at 1 y); risks: age, diabetes, BMI >40, prolonged catheter; peak first month (day 7); less after TO; UTI marks a worse course (more reoperation, de novo UUI).
- Vaginal wound infection 0.7–2%; rare: necrotising fasciitis, abscess, **adductor myositis** (may present years later).
- **Groin pain** (TO) 16–26% early → ≈4% at 12 months; NSAIDs, physio, trigger-point injection; refractory → excision — **complete removal** relieves more than partial.

### Voiding dysfunction / retention

- Voiding inefficiently but volitionally: timed voiding, biofeedback, PFMT, selective drugs before surgery.
- Tape **too tight** or **too proximal** (obstructing slings usually near bladder neck), or POP (can obstruct a normal sling ≥10 y later). More after RP (TOMUS 2.7% vs 0%). Evaluate: **temporal relation**, exam, PVR, cystoscopy, UDS selectively.

> **ALGORITHM**
> Retention/high PVR → CIC or catheter (most resolve within 2 weeks) → persistent obstruction → **transvaginal midline sling incision at 2–4 weeks** → >3 months: tape fixed → excise lateral portion → rarely formal urethrolysis

- **Urethral dilation harmful — do not perform**. 70–100% stay continent after incision; **early** release → better voiding, **less** recurrent SUI (delay main predictor).
- "Recovered" women often keep frequency/urgency — partial obstruction.

> **KEY POINT**
> **Why different timing from PVS?** Fascial obstruction often settles (wait up to 3 months). Mesh is fixed by ingrowth, rarely loosens; late release needs wider dissection and causes more recurrent SUI — release within ≈4 weeks.

### Storage symptoms, exposure, sexual function

- De novo OAB ≈6–12% (up to 15–30%), same for all MUS types. Predictors: age, DO, low capacity, high detrusor pressures. Temporal link → **partial BOO first**; **cystoscopy mandatory**. Treat: behavioural, drugs (less effective than idiopathic OAB) → revision vs SNM, PTNS, botulinum toxin.
- Vaginal exposure ≈2%; †discharge, bleeding, dyspareunia, partner discomfort. †Small: topical oestrogen ± excision; larger: excision. Urinary tract mesh: excision + repair; no immediate new mesh → *Ch 119*.
- De novo dyspareunia ≈3–14%, tracks with de novo urgency and incomplete emptying more than mesh; sling removal may relieve it. Many improve (**coital incontinence** abolished). TVT–colposuspension trial: dyspareunia 34% → 13% at 5 y.

## 13 · Adjustable slings

- **Remeex**: mesh on sutures to a suprapubic **varitensor**; adjusted postoperatively. For **complex SUI** (ISD, fixed urethra, reoperation). Cure ≈85–90%; complications 32–51%, bladder injury 7–14%, readjustment 14–43%; detrusor underactivity → retention (a quarter wanted removal). Higher BMI and pad weight predict worse results. No level 1 evidence.

## 14 · Assessing the failed or troublesome sling

> **ALGORITHM**
> Symptom after sling → define: persistent SUI vs storage vs voiding vs pain/discharge → **timeline** → exam (exposure, POP, mobility, tenderness over tape) → PVR, urine culture → **cystoscopy** → UDS selectively → tailored treatment

- Recurrent SUI, mobile urethra → repeat RP MUS or PVS; fixed/ISD → PVS, bulking, adjustable sling. Storage with temporal link → treat obstruction first. Pain/exposure → excise (complete for pain) → *Ch 119*.

## 15 · Mesh controversy and regulatory status

- **2010 FDA** safety communication (mesh for POP and SUI) based on **MAUDE** reports. Professional bodies still endorse MUS with documented counselling.
- †2019 FDA order stopping transvaginal mesh concerned **POP**, not MUS. UK (2018) paused MUS pending safeguards → *Ch 119*.

## 16 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| **ValUE** | RCT non-inferiority, 630, uncomplicated SUI: UDS vs office evaluation | Office non-inferior at 1 y | UDS may be omitted in index patient |
| **SISTEr** | RCT, 655: rectus PVS vs Burch | PVS higher cure (47% vs 38% at 2 y); more voiding dysfunction, UTI, UUI; all BOO surgery in PVS arm; persisted at 5 y | PVS more effective, more obstructive |
| **Ward–Hilton** | RCT: TVT vs open colposuspension | Equal to 5 y; TVT more bladder injury; Burch more morbidity, POP | MUS replaced Burch |
| **TOMUS** | RCT, 597: RP vs TO | 80.8% vs 77.7% (1 y, equivalent); voiding surgery 2.7% vs 0%; †more groin/neurological symptoms TO; 5 y 51% vs 43% (equivalence not met) | Either; RP slightly more durable, more voiding dysfunction |
| **OPUS** | RCT (327): POP repair ± prophylactic RP MUS in stress-continent women | UI at 12 months 27% vs 43%; NNT 6.3; more perforation, UTI, bleeding | Reduces SUI but adds risk — counsel |

# 117 Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

## 1 · Goals and principles

- P2 Vaginal prolapse repair — *QoL condition, first operation best chance, restorative/compensatory/obliterative, vaginal (less morbid) vs abdominal (more durable), apex is the cornerstone*
- Operate on grade × symptoms × bother; declining surgery is legitimate. **Success = absence of symptoms** (no bulge), which drives satisfaction more than anatomy; satisfaction tracks her own stated goals.
- Individualise to physiological age, comorbidity, prior surgery, sexual activity. **First operation is the best chance** — planes lost, recurrence rises with each re-attempt.
- Routine events (catheter, pain, constipation, UTI) are perceived as complications; preoperative education improves PGI-I without changing objective cure.
- **Consent sequence**: alternatives → purpose/limits → benefits → risks → hospital course → home course → coping with catheter.
- Obliterative contraindicated if intercourse desired; compensatory (graft) for weak tissue/failed repair. Open vs laparoscopic: equal complications, faster recovery. POP usually multicompartmental.

## 2 · Choice of procedure by compartment

- P1 Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia — *POP-Q points (A distal, B proximal, D only with cervix), DeLancey levels, operation by defect, SSLF and promontory anatomy, uterus-preserving options*

| Compartment | POP-Q point | Organ |
|---|---|---|
| Anterior | Aa / Ba | Urethrocele / cystocele |
| Middle | C / D | Cervix or cuff / enterocele |
| Posterior | Ap / Bp | Rectocele / enterocele; perineal body |

| Site | Vaginal | Abdominal |
|---|---|---|
| Aa — urethra | Anterior repair; bladder neck suspension; sling | Retropubic urethropexy |
| Ba — bladder | Anterior repair; paravaginal; colpocleisis; wedge colpectomy | Paravaginal; sacrocolpopexy |
| C — cervix/cuff | USLS; iliococcygeus; SSLF; Manchester; hysteropexy; vaginal hysterectomy; colpocleisis | Hysterectomy; USLS; sacrocolpopexy; uterine suspension |
| D — cul-de-sac | **McCall** culdoplasty | **Halban**, **Moschcowitz** culdoplasty |
| Ap — posterior | Posterior repair; site-specific repair | Colpoperineopexy |

KEY POINT

Only paravaginal repair and uterosacral suspension are done by both routes. McCall = vaginal; Halban (linear sutures) and Moschcowitz (purse-string — may angulate the ureter) = abdominal.

## 3 · Anterior compartment

- **Central** (midline pubocervical attenuation) vs **lateral/paravaginal** (detached from ATFP); most combined ± apical. **Anterior repair does not suspend the apex.**

ALGORITHM

Lithotomy → catheter → hydrodissection → midline incision apex → bladder neck → dissect epithelium off pubocervical fascia laterally to ATFP → reduce bladder → interrupted 2-0/3-0 delayed-absorbable plication, bladder neck → apex → trim judiciously → close → IV dye + **cystoscopy** → pack.

- With MUS: separate midurethral incision; with PVS: single incision to midurethra. Deep sutures may enter/kink bladder or ureter → completion cystoscopy mandatory.
- Abandoned for SUI: AUA meta-analysis failure approaching **40%**.
- Recurrence varies with **definition**: 30–46% at "POP-Q ≥2" vs 9–23% "beyond hymen"; objective recurrence often doesn't reduce satisfaction.
- Absorbable mesh: modest gain; biological grafts add little. Polypropylene vs native tissue: less recurrence/re-operation but far more graft complications; QoL, dyspareunia not improved → FDA 522 → **transvaginal synthetic POP mesh withdrawn** (US 2019) → *Ch 119*.
- Concomitant sling **halves anterior recurrence** (−55%) — larger than any graft.

| Complication | Point / management |
|---|---|
| De novo SUI | Rises with stage: 2 **41%**, early 3 **53%**, advanced 3–4 **66%**; point Ba predicts |
| Detrusor overactivity | De novo 5–7%; pre-existing resolves in up to 63% |
| Retention | Usually from sling. CIC → persistent after sling: incision/urethrolysis; hypocontractile: CIC → neuromodulation → SPC |
| Bleeding (>350 mL) | Vaginal vascularity, venous complex under pubis; bleeding = wrong plane; worse in redo |
| Bladder injury | Drain before perforating endopelvic fascia; two-layer closure; irradiated → Martius |
| Others | Ureteric obstruction, VVF, shortening, inclusion cyst, dyspareunia |

KEY POINT

**Confirming ureteric efflux**: indigo carmine (blue, IV), fluorescein 10% (green, IV), phenazopyridine (orange, oral). No efflux → ureteric catheter/retrograde or remove plication sutures. Under-hydration can delay efflux — never close without proving patency.

### Paravaginal repair

- Reattach pubocervical fascia to ATFP (White vaginal; Richardson abdominal). Non-absorbable sutures 1 cm apart: vaginal **5–7**, abdominal **4–6**; cystoscopy.
- Lateral recurrence 2–8% but central ~**22%** → two independent failures. Abdominal more durable.

- **Neither colporrhaphy nor paravaginal repair treats SUI.** Vaginal route: more serious complications (transfusion 12%, haematoma, abscess, bilateral ureteric obstruction).

## 4 · Occult SUI and the concomitant sling

- High-stage anterior POP kinks the urethra, masking SUI → "stress incontinence on prolapse reduction" (8–66% of continent women).
- Test **with prolapse reduced** (pack, swabs, speculum, pessary): none standardised; low sensitivity, high specificity.
- **Staged** argument: sling obstruction needing intervention (8.5%) ≈ intervention for SUI without sling (8.3%).
- **OPUS calculator**: 7 predictors, accuracy 0.72 vs 0.54 stress test alone. **CUPIDO** (9 predictors): urodynamics adds nothing in asymptomatic women. Six-predictor model similar.
- Subjective SUI with native repair or colpocleisis: <50% bothersome SUI → sling may be omitted. SUI symptoms but negative tests: ~30% later intervention.

KEY POINT

**Sling also helps the prolapse**: adds distal anterior support → more durable cystocele repair. **Why not everyone**: OPUS — more bladder perforation, UTI, bleeding, incomplete emptying; pelvic floor symptom scores unchanged. Weigh per patient.

ALGORITHM

Advanced anterior POP → reduction stress test ± UDS → symptomatic/occult/urodynamic SUI → offer concomitant procedure → continent, test negative → individual risk–benefit (OPUS calculator); **selective, not routine**.

## 5 · Apical compartment

The apex is the cornerstone; failing to secure it raises recurrence exponentially.

### Uterosacral ligament suspension (USLS)

- Preserves natural axis; ligament breaks at discrete points.
- **Ureter–ligament distance 4 cm (sacrum) → 0.9 cm (cervix)**; ureteric sheath → **kinking**, not ligation. Sutures near sacrum entrap S1–S4 → buttock/posterior thigh pain; S3 injury >4.6 cm cephalad of spine.
- **Safe zone**: intermediate portion, ~1 cm posterior at the level of the ischial spine; needles **lateral → medial**.

ALGORITHM

After vaginal hysterectomy: cuff stays at 3 and 9 o'clock → excise enterocele sac → pack bowel → palpate spines → Allis at 5 and 7 o'clock → **2–4** non-absorbable sutures per side → anterior repair/sling now → through apical pubocervical and rectovaginal fascia → tie → **cystoscopy after tying, before trimming** → no efflux: remove most lateral ipsilateral suture first.

- Confirm ureter lies lateral before suturing (relation shifts with prolapse degree); stents if hard to palpate. Rare: small bowel injury, pelvic abscess; voiding and SUI usually improve. Vs SSLF: less risk to pudendal/gluteal vessels, midline axis; closes cul-de-sac too. Abdominal variant: 3 permanent sutures per side (Shull).

- Apical success ~98%, anterior ~81%; ureteric kinking/injury **1–11%**. Absorbable non-inferior. Premenopausal ↑ recurrence. De novo dyspareunia possible.
- **OPTIMAL**: USLS = SSLF (~65% at 2 y, strict composite); perioperative PFMT no benefit.

### Sacrospinous ligament fixation (SSLF)

- Richter/Nichols. Extraperitoneal, preserves length; posterior or anterior access. Ligament 7–8 cm under coccygeus; pudendal bundle around spine, inferior gluteal behind.
- **Safe zone**: **1.5–3.0 cm medial to the ischial spine**, into the ligament, limited depth. Too lateral → pudendal; too medial → weak ligament + sacral nerves.

> **ALGORITHM**
> Mark apex → midline incision → enter perirectal space, rectum medial → palpate spine/ligament → retractors → 2 sutures (Capio/Miya hook) → other repairs → sutures to apex (pulley stitch if permanent) → close vagina halfway → tie apex flush (**no suture bridge**) → cystoscopy → close.

- Usually unilateral right (2 sutures 2 cm apart); bilateral/anterior approach → more midline, anterior recurrence ~7–9%. **Pulley stitch** keeps knot out of lumen.
- Cure ~85%, apical recurrence ~5%; stage 4 predicts recurrence; no defect at 6 weeks → only 3% reoperated.
- **Drawback**: axis deflected posteriorly and right → **anterior recurrence** (~18%). Concomitant anterior repair helps only when Ba > +1 (with SSLF, not USLS).
- Gluteal pain **15%**, resolves in 2–3 months (local anaesthetic injection if needed); pudendal injury; rectal injury (repair primarily); haemorrhage.

> **KEY POINT**
> Both ligaments are safest in their **middle third**, dangerous at bony ends: USLS intermediate portion; SSLF 1.5–3 cm medial to spine.

**Iliococcygeus suspension**: deficient uterosacrals; bilateral to iliococcygeus fascia **1 cm distal to the ischial spine**; anterior recurrence commonest.

### Sacrocolpopexy (SCP)

- **Indications**: SUI and other defects repairable at same sitting; failed vaginal repair; young, sexually active; isolated vault prolapse; wants most durable repair. Maximises vaginal length without distortion.
- **Type I macroporous monofilament polypropylene** + secure fixation. Fascia lata: recurrence ~1/3 vs <1/10 polypropylene.

> **ALGORITHM**
> Low lithotomy, vaginal stent/EEA sizer → sigmoid left → incise peritoneum over promontory, along **right** of rectum (right ureter) → anterior longitudinal ligament → (Halban if large enterocele) → dissect bladder off anterior vagina (6–8 cm) and rectum off posterior → Y-mesh to both walls **without entering lumen** → tension → 2–3 permanent sutures at **S1** → **retroperitonealise** → anti-incontinence procedure if indicated → cystoscopy.

- Too low (S3–4) → presacral venous plexus; too high → middle sacral vessels and **left common iliac vein** (often medial to artery).
- Tension: open — two-finger gap between graft and rectum; minimally invasive — extend vagina with sizer, then release; avoid hypersuspension (pain, de novo SUI). Absorbable mesh-

to-vagina sutures equivalent.

- Robotic: ports ≥9 cm apart, steep Trendelenburg, side-dock; extra time is suturing. Vaginal entry: two-layer closure; wrong plane ↑ exposure. Spondylodiscitis from sutures/tacks.
- Open, laparoscopic, robotic: 87–100% success; robot ~90 mL less blood loss, costlier. Learning curve: laparoscopic 30–40, robotic ~50.
- Failures mainly **posterior/distal**, essentially no apical retreatment; isolated SCP leaves cystocele/rectocele in up to one third, ~20% need vaginal repair — counsel.

| SCP complication | Rate / management |
|---|---|
| Haemorrhage (presacral) | ~4%; † pressure, sutures, bone wax, sterile thumbtacks |
| Cystotomy / enterotomy / ureter | 3% / 1.6% / 1%; bladder commonest robotic visceral injury |
| Ileus / SBO | 3.6% / 1.1% |
| Mesh exposure | **~10% by 7 years** (E-CARE); mean onset 14 months; more with microporous (Gore-Tex: bacteria enter pores macrophages can't); excision *→ Ch 119* |

- **Hysterectomy at SCP**: total ↑ exposure → **supracervical** or meticulous cuff closure.
- Cochrane vaginal apical vs SCP: more awareness (RR 2.1), repeat surgery (2.3), recurrence (1.9), SUI (1.9), dyspareunia (2.5) after vaginal — yet vaginal suits many older women. USLS = SSLF.

> **ALGORITHM**
> Elderly/comorbid, not sexually active → colpocleisis → fit, primary, wants vaginal → USLS or SSLF ± vaginal hysterectomy or hysteropexy → young/active, recurrent, short vagina, high stage → SCP (minimally invasive).

## 6 · Hysterectomy vs uterine preservation

- † Hysterectomy alone does not cure prolapse — always resuspend apex.
- With SCP, keeping the cervix lowers mesh exposure. Vaginal hysterectomy + vault suspension also effective.
- Sacrospinous hysteropexy non-inferior (newer data superior) to vaginal hysterectomy + USLS and to uterosacral hysteropexy.
- † Preservation needs a normal uterus: exclude abnormal bleeding, endometrial/cervical pathology, Lynch syndrome; cervical screening.

## 7 · Obliterative surgery — colpocleisis

- Older women not wanting coitus; regional/local anaesthesia, low morbidity and recurrence.
- **Total** (post-hysterectomy): whole vagina denuded, inverted with purse-string/transverse sutures; enterocele need not be opened (layered closure obliterates the space).
- **Le Fort**: uterus in situ, anterior/posterior rectangles excised (3 cm from meatus to 3 cm from cervix), **lateral channels** for drainage; transverse rows from leading edge to above levator plate.
- Both: **high perineorrhaphy/levator myorrhaphy**; cystoscopy. Uterus left: Pap, ultrasound ± endometrial biopsy. Hysterectomy removes cancer/**pyometra** risk but ↑ complications (OR ~1.9).

- Spare ≥**1.5 cm** distal anterior wall (urethral traction SUI); sling if SUI/occult SUI; frail → consider SPC.
- Success ~90–98%, satisfaction >90%, regret up to **13%**; 43% >80 years — optimisation matters most.

## 8 · Posterior compartment

- Symptoms: bulge, splinting/digitation, trapping (**not faecal incontinence**), dyspareunia. Exclude slow transit and **anismus** (biofeedback) first.
- **Predicts benefit**: bulge, splinting, incomplete emptying on defecography, large rectocele. Not: constipation. **Constipation often persists**; two-thirds of digitators may resume.
- **Transvaginal beats transanal** (recurrence 7% vs 66%).
- **Midline fascial plication** (avoid wide lateral bites) ± perineal body = ICI procedure of choice. **Site-specific**: commonest defect **transverse** (detached from perineal body); higher recurrence in advanced POP. Perineorrhaphy: triangular excision, mattress approximation of perineal body, smooth contour (ridges → dyspareunia).
- Plication steps: repair anterior/apical first → hydrodissection → midline incision → dissect rectovaginal fascia off epithelium → purse-string at hernia base if large → midline 2-0 absorbable plication → incorporate perineal body → trim, close.
- Plication relieves bulge, straining, splinting; satisfaction follows defecatory relief more than POP-Q. Cochrane: evidence insufficient.
- **Levator plication abandoned** — dyspareunia up to 50%. **Grafts** not supported (complications RR 1.8).

## 9 · Complications across POP surgery — summary

| Problem | Prevention / management |
|---|---|
| Ureteric injury (USLS highest, 1–11%) | Lateral → medial bites; cystoscopy after tying; remove most lateral suture; stent; reimplant |
| Bladder/rectal injury | Correct plane, drained bladder, rectal finger; layered closure |
| Mesh exposure | Type I mesh, full-thickness flaps, supracervical hysterectomy; oestrogen/excision |
| Recurrence | Anterior after USLS/SSLF, posterior after SCP; most asymptomatic |
| Buttock pain | Middle-third placement; observe ≤3 months; nerve injection |
| De novo SUI / retention | Reduction testing; selective sling; CIC; sling incision |

## 10 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| CARE | RCT: SCP ± Burch, continent, stage 2–4 | SUI 24% vs 44% at 3 months | Prophylactic Burch reduces de novo SUI |
| E-CARE | 7-year extension | SUI 62% vs 77%; mesh exposure ~10%; composite failure 34–48% | Benefit = delay; exposure rises with time |
| OPUS | RCT: vaginal POP repair ± MUS vs sham | 27% vs 43% at 12 months; more perforation, UTI, bleeding, retention | Selective sling |

| Trial | Design | Result | Take-home |
|---|---|---|---|
| OPTIMAL | RCT: USLS vs SSLF ± PFMT | Equal (~65%) at 2 y; PFMT no effect | Native apical repairs equivalent |
| Bilateral SSLF vs abdominal SCP | RCT | Earlier failure, 3× cystocele re-operation after SSLF | Abdominal more durable |
| Transvaginal vs transanal rectocele | RCT | Recurrence 7% vs 66% | Repair vaginally |

# 119 Genitourinary Complications of Mesh and Their Repair

## 1 · Mesh, terminology and regulation

- Pelvic floor ≠ abdominal wall: near viscera, large stretch, abundant microbiota. Harms: pain, exposure/extrusion, injury to nerves, muscle, viscera, vessels.

| Amid type | Pore | Example | Behaviour |
|---|---|---|---|
| **I** | Macroporous >75 µm | Knitted monofilament polypropylene | **Only recommended mesh** |
| II | Microporous ~10 µm | PTFE (Gore-Tex) | Encapsulated; >30% full-length slings excised |
| III | Macropores in microporous base | Mersilene, Mycromesh | Extrusion ~20% |
| IV | Submicronic | Silicone-coated | Extrusion/sinus up to 40% |

KEY POINT

**Why pore size matters**: bacteria (~1 µm) enter pores that macrophages (~50 µm) cannot — infection sealed from immunity. Large pores (~1500 µm) allow fibrocollagenous integration. Woven polypropylene (Marlex) is stiff and extrudes. No excess cancer or autoimmune disease with mesh.

- P1 Management of perforated mid urethral sling — *IUGA/ICS terms: exposure, extrusion, perforation; "erosion" not used; trocar injury and late perforation management*
- Also: contraction (shrinkage), prominence (wrinkling/folding without epithelial defect).

† **CTS coding**: Category 1–3 vaginal (1 no separation; 2 exposure ≤1 cm; 3 >1 cm or extrusion), 4 urinary tract, 5 bowel, 6 skin/musculoskeletal, 7 systemic; suffix A asymptomatic, B symptomatic, C infection, D abscess. Time T1 ≤48 h, T2 48 h–2 months, T3 2–12 months, T4 >12 months. Site S1 vaginal suture line, S2 vagina away from it, S3 trocar path, S4 other skin/musculoskeletal, S5 intra-abdominal.

- **FDA**: class I low, II intermediate, III high risk (premarket approval). Male/female SUI mesh and **transabdominal POP = class II**. Transvaginal POP mesh: warnings 2008/2011 → **class III (2016)** → sales stopped **2019**; not slings or sacrocolpopexy.
- Multi-incision MUS and sacrocolpopexy: safety established; mini-slings not adequately demonstrated.
- Written mesh information; standard consent for removal; policy for returning explanted mesh.

## 2 · Risk factors and prevention

- **Contraindications to vaginal mesh**: simultaneous urethrovaginal fistula repair, urethral injury/mesh perforation, diverticulectomy. **Intraoperative urethral injury → abandon mesh**, sling later.
- Relative: prior fistula/diverticulum surgery, radiation, poor/oestrogen-deficient tissue, smoking, obesity, age extremes; men: radiotherapy, urethrotomy.

- POP mesh: high mesh volume, **shallow vaginal dissection**, hysterectomy, diabetes, sexual activity.

## 3 · Evaluation

> **ALGORITHM**
> History + **prior operative notes** → speculum/pelvic exam (exposure, bands, point tenderness, levator hypertonicity) → cystourethroscopy if haematuria, recurrent UTI, storage LUTS, voiding pain → UDS if voiding dysfunction → translabial ultrasound for sling; **MRI** for sacrocolpopexy mesh (invisible on CT); CT for haematoma.

Time since surgery never excludes a complication — slings can expose decades later.

## 4 · Midurethral sling complications

### Vaginal exposure

- Causes: infection, haematoma, excess tension, superficial placement. Bleeding, discharge, dyspareunia, hispareunia (partner pain), UTI.
- Small early exposure: **topical oestrogen** (1 g cream nightly 2 weeks then 2–3×/week; or estradiol 10 µg tablet).
- **Excision**: hydrodissect → incision or inverted-U flap → locate (sound, ultrasound) → expose full width → pass instrument behind, divide → excise ≥ **1 cm** around defect → avoid blind lateral dissection (vulval haematoma, esp. transobturator) → close.

### Pain

- Risk: young age, fibromyalgia, prior pelvic pain. Causes: shrinkage, tension, obturator neuralgia, hypertonia.
- **Immediate severe pain**: exclude haematoma, consider **early removal** (before ingrowth).
- Delayed focal: PFPT, trigger point injection, oestrogen → excise painful segment (60–90% improve). Diffuse late pain → central sensitisation; excision may fail.
- Remove **as much vaginal mesh as possible** (81% pain-free). TO **groin pain** → vaginal + thigh incision through adductor brevis to obturator externus.

### Bladder perforation

- Unrecognised trocar puncture (low BMI, prior surgery); **full bladder** at cystoscopy. Storage LUTS, haematuria, UTI, stones, VVF.
- **Endoscopic** (holmium, scissors, resectoscope): low morbidity, mesh stays in wall — re-extrusion. **Open/robotic transvesical**: excise 2 cm beyond defect, two-layer closure, catheter 2–3 weeks; minimal re-extrusion.

### Urethral perforation

- From trocar injury, over-tensioning or **urethral dilation** (avoid — divide obstructing sling). Voiding pain, retention, haematuria.
- Definitive: inverted-U flap → divide uninjured arm → peel mesh off → minimal urethrotomy → layered closure ± **Martius** → non-overlapping suture lines; stricture → buccal urethroplasty.
- After excision: SUI **42%** (fewer with simultaneous fascial sling).

### Voiding dysfunction

- Midline **sling division**: retention resolves 95%, symptoms 75%; urgency incontinence persists up to 76%, de novo up to 45%. Recurrent SUI 20–74% → bulking, repeat MUS, fascial sling.
- Persistent: UDS (obstruction vs underactivity vs dysfunctional voiding) → wider excision, urethrolysis. Post-excision haematoma: expectant unless expanding/unstable.

> **KEY POINT**
> **How much to remove**: retention — midline division; exposure — ≥1 cm each side; pain — maximal vaginal removal, rarely arms. **Never remove asymptomatic mesh**; more removal = more bleeding, POP and SUI recurrence.

## 5 · Transvaginal POP mesh

- Largest RCT (n = 1358): native vs mesh equal at 6 y (re-operation 12% vs 10%); 8% of mesh arm needed excision.
- Kits (Perigee, Apogee): arm routes differ — get the operative note.
- Exposure 10–12%, removal 5–8% (up to 8 y); **contraction** ~12% (unique to POP mesh; tightness, dyspareunia); bladder perforation 0.7–3.5%, rectal 0.15%, fistula 0.2%.
- **Conservative** if <1 cm, minimal symptoms, no infection: observe or oestrogen; only **size** predicts success. Office trimming: 73% later need theatre.
- **Excision**: EUA, cystoscopy/proctoscopy → thick flaps → clamp behind mesh and divide (**most dangerous step** — most visceral injuries) → cut arms under tension (don't chase) → native repair.
- Sequelae: cystocele recurrence 19%, repeat excision 20%, VVF 3%; multicompartment excision ↑ transfusion (1% → 9%); 45% abstain sexually.

## 6 · Sacrocolpopexy mesh

- Exposure ~**10.5% at 7 years**; total hysterectomy 23% vs 5%; stage 3–4. Cervix-sparing low risk.
- Discharge, granulation, dyspareunia; cuff abscess; bladder/bowel perforation; **discitis**. Needs cross-sectional imaging.
- **Exposed suture**: cut below knot in office. Small exposure: oestrogen 6–8 weeks (low success). Vaginal excision often fails; partial colpocleisis ~50% success, shortens vagina — not if sexually active.
- **Abdominal removal** (failed vaginal, pain, infection, discitis, ureteric complication): **remove entire infected mesh** incl. sacral sutures — residual mesh causes sinus, abscess, fistula. Partial vaginectomy often (74%); bowel injury up to 29%.

## 7 · Male slings

- Fewer complications (clean wound, deep cover, thick bulbar urethra).
- **Advance XP**: retention 14–21%, usually days; incise one arm if persistent; neobladder highest risk. **Virtue**: perineal/scrotal pain 14–17%, chronic 0–7%. InVance: divide before AUS.
- **Urethral extrusion** (usually unrecognised perforation without intraoperative cystoscopy): urethroscopy in persistent retention; excise lateral to urethra, primary or buccal repair; avoid transection.

# 121 Urinary Tract Fistulae

## 1 · Definition and universal principles

- **Urinary tract fistula**: abnormal epithelial- or mesothelial-lined connection between urinary tract and another cavity (bowel, vessel, genital tract) or skin.
- Causes: iatrogenic (commonest in developed world), malignancy, inflammation, infection, ischaemia, trauma, parturition.
- Persists because **urine keeps flowing through a tract that cannot heal** → tract epithelialises. Every principle diverts flow or restores healing.
- The **first repair has the best chance** — optimise patient and surgeon first.

| Management (whole course) — Table 121.1 | Operation (any route) — Table 121.2 |
|---|---|
| Optimise nutrition | Evaluate concomitant injury; rule out malignancy |
| Control infection | Adequate exposure; remove all foreign material |
| Continuous drainage (catheter/stent) | Tension-free closure; vascularised edges |
| Eliminate obstruction | Multiple, watertight, **non-overlapping** layers |
| Consider malignancy — **always biopsy** if any history | Consider interposition; drain after |

**KEY POINT**

Trap: a fistula may be the only sign of recurrent cancer — biopsy the tract after prior malignancy or radiation.

## 2 · Vesicovaginal fistula (VVF) — aetiology

Commonest acquired urinary fistula (**>75%** of genitourinary [GU] fistulae).

| | Developing world — obstetric (>90%) | Developed world — iatrogenic |
|---|---|---|
| Cause | Obstructed labour; young, small pelvis, poor nutrition | Unrecognised surgical injury; **>75%** after hysterectomy |
| Mechanism | Fetal head impacted for days → pressure necrosis → large defect | Dissection, cautery, clamp crush, bladder caught in cuff closure |
| Extent | Trigone, urethra, ureter often involved; "obstructed labour injury complex" (renal, rectal, orthopaedic, neurological, psychosocial) | Solitary, **supratrigonal at vaginal cuff** |

- **Hysterectomy**: VVF ≈ 0.8/1000; lowest vaginal, highest laparoscopic (highest 30-day rate after LAVH). Risk ↑ with distorted anatomy (endometriosis, fibroids, prior surgery/radiation, malignancy, bleeding) and concomitant prolapse/incontinence/mesh surgery. Immediate recognition and repair prevents fistula.
- **Universal cystoscopy at hysterectomy**: ↑ intraoperative detection; effect on delayed injury inconsistent; adds time and UTI. AUGS: routine with any prolapse/anti-incontinence procedure; AAGL: routine after laparoscopic/robotic hysterectomy; ACOG: no recommendation after benign hysterectomy.
- **Radiation**: obliterative endarteritis + fibrosis → necrosis. 0.6–2% after cervical cancer radiotherapy; typically **1.5–2 years** later (up to 30); dose-dependent, worse in smokers.
- Also mesh (sustained inflammation), **bevacizumab** (anti-VEGF, hypoxia), cancer; rare: pessary, IUD, infection, lasers, endometriosis.

## 3 · Classification of VVF

>25 systems; WHO: simple vs complex.

| Axis | Goh | Waaldijk |
|---|---|---|
| Basis | Distance of distal edge from external urethral meatus | Closing (continence) mechanism and urethral involvement |
| Types | 1: >3.5 cm · 2: 2.5–3.5 · 3: 1.5–2.5 · 4: <1.5 cm | I: closing mechanism spared · IIA: no (sub)total urethral involvement · IIB: (sub)total urethral involvement (A/B = without/with circumferential defect) · III: miscellaneous (ureteric) |
| Size | a <1.5 · b 1.5–3 · c >3 cm | Small <2, medium 2–3, large 4–5, extensive ≥6 cm |
| Tissue | i none/mild fibrosis · ii moderate/severe · iii special (radiation, ureteric, circumferential, previous repair) | — |

**KEY POINT**

Trap: Goh type 1 is farthest from the meatus (best), type 4 closest (worst) — closer = worse closure and continence. Goh predicts closure better. Always describe size, location (trigonal vs supratrigonal), tissue loss, ureteric and urethral involvement.

## 4 · Diagnosis of VVF

- Classic: **continuous leakage day and night**, position-independent; large tract → no desire to void. Small → positional leak only. Also recurrent UTI, haematuria, "discharge".
- **Timing points to mechanism**: missed cystotomy → first days; thermal/crush → **7–10 days**; radiation → years. Fever, flank pain, ileus → urinoma/ureteric injury.
- Speculum (apex commonest post-hysterectomy); vaginal fluid creatinine ≈ urine creatinine. Biopsy friable tissue if prior cancer/radiation; EUA if stenosed; vaginal oestrogen for atrophy. Optimise smoking, glucose, nutrition.
- **Single dye**: backfill ~300 mL methylene blue → tampon → walk 15–60 min. Proximal stain → VVF; distal → urethrovaginal fistula or SUI; from cervical os → vesicouterine fistula.
- **Double dye**: oral phenazopyridine 100–200 mg (orange in 1 h) + bladder blue → tampon: **orange = UVF**, blue = VVF, both = both.
- **Cystoscopy is integral** (sensitivity ~92%): size, number, orifices, bladder neck, mesh, capacity, biopsy; probe small tracts with guidewire/5-Fr catheter; under anaesthesia allows retrograde pyelography and stenting. **Vaginoscopy** for vaginal side, biopsy, radiation stenosis.
- Cystogram/VCUG: vaginal filling, best on lateral view; shows urethrovaginal fistula. MRI (T2) for difficult cases.
- **Upper tract imaging mandatory** — ureteric injury in up to **12%**. CTU with delayed phase (MR urography if contrast allergy/renal failure). **Partial ureteric obstruction with VVF = UVF until proven otherwise.** Unclear → retrograde pyelogram + stent; fails → PCN.
- UDS/VUDS to plan: 10–15% have SUI (autologous sling); radiation-reduced capacity/compliance (augmentation/diversion). Occlude a large fistula with Foley balloon/pack to fill.

**KEY POINT**

Work-up order: is it urine (creatinine), where from (dye tests, endoscopy), what else is damaged (hence compulsory upper tract imaging). A preoperative PCN also allows antegrade ureteroscopy to find the ureter in scar.

## 5 · VVF — timing, conservative care, repair

- P3 P2 P1 VVF management options; early vs delayed repair; eponymous techniques — *catheter trial (~15%, <1 cm, within 3 weeks, EAU ≥12 weeks), fulguration, sealants not recommended; timing by cause; vaginal vs abdominal route (~90% either); flap-splitting, Latzko, cuff excision, O'Conor, transvesical/extravesical, lap/robotic; flaps; postoperative care; diversion; outcomes*

| Setting | Timing of repair | Rationale |
|---|---|---|
| Obstetric (traditional) | ≥3 months, catheter meanwhile | Inflammation settles, necrosis sloughs |
| Iatrogenic, clean tissue | Early; EAU: no minimum delay if no infection or recent radiation | Comparable closure; less distress |
| Radiation-induced | ≥ **12 months** after radiation ends | Late injury still evolving |

- No consensus definition of success; anatomical closure ≠ continence. Radiation: strong failure predictor, with ureteric stricture, poor compliance, ISD — consider diversion.
- **Size**: <3 mm best spontaneous closure; <10 mm may close with drainage; ≈ >3 cm (≥1.5 cm for residual obstetric incontinence) predicts failure.
- **Catheter**: favourable <1 cm, within **3 weeks**; unfavourable ≥6 weeks, radiation, malignancy (never heal). EAU trial ≥ **12 weeks**. Sealants possible for <1 cm in poor surgical candidates (catheter adherence, spasm, stone on glue).
- **Location**: within **1 cm** of ureter → stent; ureter involved → reimplant (abdominal); bladder neck → reconstruction ± pubovaginal sling; apical (8–10 cm from introitus) beyond Martius reach → peritoneal/omental flap.
- Route: no RCTs; first-attempt closure ≈ **90%** either. Transvaginal (less blood loss, pain, stay) — some call gold standard. Abdominal: large (≥4 cm primary, ≥2 cm recurrent), urethral/bladder-neck/ureteric involvement, non-capacious vagina, radiation, cannot be placed in lithotomy; vaginal suits dense abdominal adhesions, no GA. Vaginal risks shortening/dyspareunia (esp. Latzko); abdominal leaves sexual function unchanged.
- Tract excision and interposition not proven in simple VVF; valued if irradiated, recurrent, poor tissue.

**ALGORITHM**

**Transvaginal flap-splitting (Table 121.5)**: cystoscopy (stent close ureters; wire through tract) → SPC + urethral catheter → episiotomy if narrow → Foley ≥8 Fr through fistula for traction → hydrodissect → **J/U-incision** with 1–3 mm cuff → wide anterior/posterior flaps → layer 1: 3-0/4-0 delayed absorbable → layer 2 **perpendicular** → leak test 200–300 mL → ± Martius/peritoneal flap → advance opposite flap ≥2 cm beyond (layer 3) → cystoscopy for efflux

- **Postoperative drainage is the most important part**: urethral ± SPC ≥10–14 days (two catheters guard against blockage); ± cystogram; spasms → anticholinergic/β3-agonist; drain creatinine for urinoma; stents out 4–6 weeks; †pelvic rest ~6 weeks.

- **Latzko**: bladder not opened, tract not excised; denude 1–2 cm, close in 2+ superficial layers; short, no ureteric dissection; ≥90% non-irradiated; favoured in radiation/orifices close; drawback shortening.

ALGORITHM

**O'Conor (Table 121.7)**: ureteric catheters → low midline, extraperitoneal → sagittal dome cystotomy → **bivalve** down around fistula → excise tract → separate bladder from vagina ≥2–3 cm beyond → close vagina → **omentum/peritoneum** beyond vaginal line → close bladder 1–2 layers → SPC ± urethral, fill test → 10–14 days, cystogram

- O'Conor failure: carcinoma ± radiation, wound infection, poor mobilisation; more de novo UUI and LUTS (long suture lines). Transvesical (limited cystotomy): restricted mobilisation/interposition. Extravesical: orifices not seen (ureteric risk), scarred plane, infratrigonal hard; suits supratrigonal (transvesical suits low-lying).
- Robotic O'Conor: close vagina **transversely**, bladder **vertically** (lines cross); omental J-flap/epiploic appendage/peritoneum; ICG for edge vascularity. Contraindication: dense adhesions.

## 6 · Interposition flaps

- Reinforce, fill dead space, import blood/lymphatics. For **complex VVF** (radiation, recurrent, ischaemic obstetric, poor tissue); RCT: **no benefit of Martius** in simple VVF.
- **Martius**: supply internal pudendal (inferior), external pudendal (superior), obturator (lateral). Labial incision 4–7 cm; mobilise lateral to bulbocavernosus; keep **superior** pedicle for VVF (inferior for rectovaginal); tunnel ~2 cm; labial drain 48 h. Reaches mid-distal VVF, bladder neck, urethra — **not apex**.
- Vaginal peritoneal flap (pouch of Douglas) for apex; **omentum** (gastroepiploic) for radiation/recurrent/large; abdominal peritoneum or sigmoid epiploic appendages if omentum short; rectus, gracilis, Singapore — last resort.
- Outcomes: residual UUI/SUI (Goh 3–4, >1 repair, positive cough/pad test) drives QoL and depression; ICS: physiotherapy for all obstetric fistula. Dyspareunia: small vagina, fistula ≥3 cm. Diversion if unrepairable (resource-poor: ureterosigmoidostomy).

## 7 · Ureterovaginal fistula (UVF)

- P3 Urological complications in OBS-GYN and uretero-vaginal fistula — *distal ureteric injury at hysterectomy, danger sites, normal voiding + leak within 4 weeks, double dye/CTU/RGP, stent within 2 weeks for 4–6 weeks, reimplant ± psoas hitch/Boari, ≥90%, follow 30 months*
- Up to **25%** of VVF have a coexisting UVF; delayed recognition ~4-fold fistula risk. Preoperative IVP does not help; prophylactic stents selective (e.g. caesarean hysterectomy for placenta accreta). Ileal ureter/nephropexy for length; early vs ≥6-week repair equal.

ALGORITHM

Suspect UVF → exclude VVF + CTU/retrograde pyelogram → stent → remove 4–6 weeks → reimage → resolved, or persistent/stent fails → ureteroneocystostomy

KEY POINT

UVF: wet but voids normally; the ureter heals over a stent if reached early, so detection-to-stent interval largely decides whether surgery is needed.

## 8 · Vesicouterine fistula (VUF)

- Mostly after **low transverse caesarean**; uterine body above internal os ↔ posterior bladder; competent cervix → **continence preserved**.
- **Youssef syndrome**: menouria, amenorrhoea, first-trimester miscarriage, continence. Incontinent only if tract below os or cervix incompetent. Differential of menouria: bladder endometriosis.
- Diagnosis: dye from cervical os; cystoscopy (posterior supratrigonal defect; scope may enter uterus); cystogram, hysterosalpingography, MRI; image upper tract.
- Spontaneous closure ~5%. Wait 3–4 months (involution). Transabdominal separation, layered closure + omentum/peritoneum (uterus-sparing) or hysterectomy if family complete; Martius rarely reaches. After: contraception until healed, caesarean delivery.

## 9 · Urethrovaginal fistula

- P3 Urological complications of OBGY surgery and urethrovaginal fistula — *diverticulectomy commonest Western cause (20–25%), distal vs proximal symptoms, flexible urethroscopy, ~20% separate VVF, flap repair + Martius, autologous sling never synthetic, SPC ≥14 days, closure 90–95% vs continence 50–80%*
- Hispareunia after mesh.

## 10 · Rectourethral fistula (RUF)

- P3 Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management — *post-prostate-cancer treatment, faecaluria/pneumaturia, biopsy, colostomy first, transperineal gracilis vs York-Mason vs abdominal, salvage diversion*

| Martini | Category |
|---|---|
| Stage (size) | I ≤1.5 cm · II >1.5 cm · III any size with sphincter damage |
| Grade (tissue) | 0 surgical injury only · 1 prior radiation/energy · 2 salvage RP or salvage ablation |

## 11 · Vesicoenteric fistula (VEF)

- Colovesical commonest; M : F **3 : 1** (uterus shields; prior hysterectomy common in affected women).
- **Diverticulitis** > **Crohn** (ileovesical) > colorectal cancer (10–15% of colovesical have neoplasm), bladder/prostate cancer, radiation, appendicitis. †Sigmoid → left posterolateral bladder/dome.
- **Pneumaturia** (50–70%), faecaluria, polymicrobial UTI. **Gouverneur syndrome**: suprapubic pain, frequency, dysuria, tenesmus.

| Test | Role |
|---|---|
| Poppy seed test | Highest detection of presence; cheap; no location |
| **CT** (preferred) | Air in bladder, wall thickening, adherent bowel; shows cause |
| Cystoscopy | Nonspecific erythema/oedema |
| Colonoscopy | Poor for tract; **excludes cancer** |
| MRI | Crohn, anatomy |

- Conservative if minimal symptoms, benign (esp. Crohn — immunomodulators, nutrition); spontaneous closure <2%.

- **Resect diseased bowel with fistula**: single-stage anastomosis (± loop ileostomy), increasingly laparoscopic; Hartmann for abscess, contamination, advanced cancer, radiation. Bladder: leave/oversew small defect; **partial cystectomy** if malignant. Interpose omentum.

> **KEY POINT**
> VEF is a bowel disease presenting in the bladder — remove the diseased bowel; the bladder usually heals itself.

**Renocolic**: spontaneous mostly renal (XGP, chronic infection) → nephrectomy ± bowel resection; small iatrogenic (PCNL colonic injury) → antibiotics, stent/nephrostomy, nil by mouth.

## 12 · Urocutaneous fistulae

- **Vesicocutaneous**: after long-term SPC, open prostatectomy, radiation. Persists with epithelialised tract, infection, foreign body, tumour, **BOO**. Drain, catheterise, relieve BOO; fistulectomy/partial cystectomy if non-healing.
- **Urethrocutaneous**: commonest hypospadias complication (up to 20%). Wait ≥**6 months**; exclude distal stricture. <1 cm → local closure; ≥1 cm/redo → vascularised flaps; **tunica vaginalis** > dartos; massive loss → gracilis; neurogenic with pressure ulcers → ileal conduit.
- **Nephrocutaneous**: stones, XGP, prior surgery, TB (≥10%); non-functioning in ~90% → **nephrectomy**; contrast CT.

## 13 · Pubovesical fistula (PVF)

- Bladder/prostate/urethra → pubic symphysis → **osteomyelitis**. Prostate radiotherapy + repeated outlet procedures (TURP, BNI, stricture); median ~10 years after radiation; also after AUS.
- **Pubic pain on walking**; abscess, urosepsis, sinus. **MRI most sensitive and specific**; CT-guided bone biopsy (Candida, Enterococcus, Pseudomonas; urine culture unreliable).
- Conservative usually fails. Debride bone, excise fistula, divert, rectus/omental flap; antibiotics 4–6 weeks. Primary repair only if sphincter functional, bladder normal, no distal obstruction; else **cystoprostatectomy + diversion**.

## 14 · Urovascular fistulae

- **Arteriocalyceal**: after PCNL, transplant, trauma; ± pseudoaneurysm/AVF → **angioembolisation**.
- **Ureteroarterial (UAF)**: emergency. Secondary (85%): long-term stents, pelvic surgery, radiation, cancer, vascular grafts; common iliac commonest. **Intermittent haematuria**; ~20% unstable.
- **Angiography** (provocative stent removal only with endovascular treatment ready); cystoscopy: pulsatile bleeding from one orifice; CTA insensitive; negative test does not exclude.
- **Endovascular stent graft** ± internal iliac embolisation; open (bypass, ligation, nephrectomy) if infection/abscess/enteric involvement.

## 15 · Renal urine leak and anastomotic leak

- **Trauma**: leak in 1–7% (~quarter of grade IV–V). CT with 10-min delayed phase; non-opacified distal ureter → suspect UPJ avulsion → retrograde pyelogram. Most resolve; persistent →

stent/nephrostomy.

- **Partial nephrectomy** (0.5–20%; low robotic): central tumour predicts. Drain creatinine, CT; drain ± stent/nephrostomy; refractory → glue, embolisation.
- **Vesicourethral anastomotic leak after RP** — Patil I extraperitoneal <6 cm, II extraperitoneal ≥6 cm, III intraperitoneal. Prolonged catheter → withdraw drain → stents/nephrostomies → reoperation last.

## 16 · Fistula after gender-affirming surgery

- Transmasculine (phalloplasty, metoidioplasty): urethrocutaneous fistula and stricture common (~3 months); **primary colpectomy** roughly halves fistula rate. Small → catheter; persistent → excision/regrafting.
- Transfeminine: urethro-/vesiconeovaginal fistula rare → suprapubic diversion → repair with interposition.

## 17 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| Martius flap in simple obstetric VVF | RCT, repair ± Martius | No difference in closure; ~⅓ donor-site pain/numbness | Reserve Martius for complex fistula |
| Prophylactic ureteric stents in gynaecological surgery | RCT (+ meta-analysis) | No/small reduction in injury (NNT 224) | Selective, not routine, stenting |

# 122 Bladder and Female Urethral Diverticula

## 1 · Bladder diverticula — definition and pathology

- P2 Bladder diverticulum. — *definition, congenital vs acquired, investigations, management, diverticulectomy steps, T3a staging*
- Urothelium herniating through muscularis propria; wall = mucosa, lamina propria, scattered non-functional muscle, adventitia; outer fibrous pseudocapsule gives the surgical plane.
- **No functional muscle → sac cannot contract** → high PVR, stasis → infection, stones, malignancy. Neck ringed by middle-detrusor fibres → narrow ostium (target of endoscopic neck resection).
- ~90% in adults; M : F ≈ **9 : 1**. Cellules → saccules → diverticula: a pressure continuum.

| Feature | Congenital | Acquired |
|---|---|---|
| Age / number | <10 years; solitary | Men >60; often multiple |
| Site | Posterolateral to ureteric orifice (~90% near UVJ); VUR common | Ureterovesical hiatus commonest; anywhere |
| Cause | Detrusor weakness at UVJ | **BOO** or neurogenic — almost always |
| Wall / presentation | Smooth; acute UTI | Trabeculated; incidental (LUTS, haematuria, UTI) |
| Malignancy | Not increased | Increased |

- Congenital: up to 60% have syndrome, neuropathic bladder or BOO — **Menkes** (copper, kinky hair), **Williams**, **Ehlers-Danlos**, fetal alcohol, prune-belly, PUV. No chromosomal testing for a single simple lesion.
- Acquired: ~**70%** BPH; also bladder neck obstruction, stricture. Women: dysfunctional voiding, prolapse, primary bladder neck obstruction, post-sling obstruction.
- **Hutch diverticulum**: paraureteric, neuropathic bladder with VUR (paraplegics).

> **KEY POINT**
> Congenital = weak wall at normal pressure; acquired = normal wall at high pressure — an acquired diverticulum sends the search back to the outlet.

## 2 · Bladder diverticula — evaluation

- Nonspecific (incomplete emptying, double voiding); rare iliac vein compression, inguinal hernia content. Persistent pyuria despite antibiotics may justify excision.
- DRE, PSA, urinalysis, culture, **cytology**, endoscopy, upper and lower tract imaging; seek BOO or occult neuropathy.
- **VCUG** (AP, oblique, lateral): size, VUR (up to 13% congenital), emptying. **Paradoxic enlargement**: sac fills on voiding, refills bladder after.
- CT/MRI: number, masses, relation to ureters/rectum; needed if neck obstructed.
- HUN in ~7% adults (upper tract abnormality up to 30% children): compression, underlying dysfunction, VUR. Pelvic ureter deviated **medially**; ureter entering a sac loses backing → reflux.
- **VUDS**: **pressure-sink effect** — contractility falsely low; raised PVR may be diverticular urine.

- **Cystoscopy**: inspect whole sac (flexible), cytology from sac; biopsy with extreme care (perforation seeds tumour). CT can miss tumour.
- VUR with congenital diverticulum: reimplant **selectively** (scarring, deteriorating function). Untreated cause → recurrence and postoperative voiding problems; treating it may make excision unnecessary.

> **KEY POINT**
> Two jobs: find the cause (outlet or neurogenic) and exclude cancer in a sac each test sees imperfectly — so tests are combined.

## 3 · Malignancy in bladder diverticula

- Stasis + inflammation; prevalence 0.8–10%, peak 65–75 years. Urothelial **70–80%**, SCC 20–25%.
- **No muscularis propria** → early extravesical spread, perforation/seeding at TUR, inaccurate staging; clinical stage key prognostic factor; upstaging common.
- **Radical cystectomy** — same stage-for-stage survival; partial cystectomy/diverticulectomy for selected low-grade, low-stage; some favour open excision without prior TUR; TUR + intravesical role undefined.

> **KEY POINT**
> Treat a diverticular tumour as potentially invasive from the outset — missing muscle explains early spread, perforation risk and inaccurate staging.

## 4 · Bladder diverticula — management

> **ALGORITHM**
> Diverticulum → history, cystoscopy, cytology, VCUG → suspicious lesion: biopsy → cancer therapy; none: upper tract imaging → HUN or symptoms → urodynamics → treat BOO → reassess → persistent problem from diverticulum → fit: diverticulectomy; unfit: CIC → asymptomatic, uncomplicated: surveillance

- **Diverticulectomy indications**: refractory LUTS, relapsing UTI, **stones**, **carcinoma/premalignancy**, upper tract deterioration (obstruction, VUR). **Size alone is not an indication.**
- **Treat outlet first or simultaneously**; small sac may then empty (TURP → reassess → surveillance). Concomitant prostatectomy common.
- Observation: counsel on cancer; surveillance with symptoms, cytology, cystoscopy (no fixed schedule; natural history unknown). CIC/indwelling if poor emptying in unfit.
- **Endoscopic** (unfit/at TURP): resect neck to muscle ± fulguration of lining; risk retention from "venting"; poor for large sacs.
- Diverticulectomy open/lap/robotic *→ Ch 124*; †ipsilateral ureteric catheter, intra-/extravesical dissection in pseudocapsule, layered neck closure, reimplant ureter if it enters sac.
- Syndromic multiple diverticula in children: observe (poor healing, recurrence).

## 5 · Female urethral diverticulum (UD) — anatomy and aetiology

- P1 Female urethral diverticula — *Routh theory, posterolateral ostium, three Ds ~5%, stones 4–10%, adenocarcinoma, imaging sensitivities (MRI 97%), L/N/S/C3, diverticulectomy principles, autologous sling only, recurrence 10–22%, de novo SUI 7–16%*

- Urethra 3–4 cm; urethropelvic ligament (abdominal leaf = endopelvic fascia; vaginal leaf = **periurethral fascia**); UD lie between leaves.
- **Periurethral glands** posterolateral, distal two-thirds; ducts drain into **distal third**. Skene glands largest, most distal, lateral to meatus.
- Striated sphincter horseshoe, mid-urethra, deficient vaginally — UD overlaps it, plan continence. Proximal urethra: urothelium, iliac nodes; distal: squamous, inguinal nodes.
- **Young–Wahle**: gland abscess → disrupts muscle → herniates into fascia, usually **ventrally** → ruptures into lumen. Routh: E. coli, gonococcus.
- Against gland theory: ~⅓ proximal. Bulking agents mimic UD; mesh slings do not cause UD. Found in proportion to how hard sought.
- Configuration: simple ~73%, saddlebag ~17%, circumferential ~10%; mid-urethral ~50%, proximal/distal ~25% each; multiple ~6–19%.
- Carcinoma in UD: **adenocarcinoma** > urothelial > SCC (primary female urethral cancer: SCC). Imaging misses small cancers — biopsy if non-excisional therapy. Local excision alone recurs: anterior exenteration/cystourethrectomy ± RT, or excision + RT/chemo. Nephrogenic adenoma mimics adenocarcinoma.

## 6 · UD — presentation and evaluation

- All three Ds in only ~**5%**. Irritative LUTS, pain, recurrent UTI; mass (~⅓), incontinence (~⅓), haematuria, discharge, retention; 10–20% asymptomatic. **Size does not track symptoms**; misdiagnosed for years (IC, vulvodynia) — examine.
- Pregnancy: antibiotics/aspiration, repair after delivery.
- Mass 1–3 cm inside introitus; meatal distortion → Skene; **hard → stone or cancer**. Atrophy (oestrogen), stress test, prolapse. Sterile culture does not exclude infection.
- **Cystourethroscopy**: ostium at **4 and 8 o'clock** mid-urethra; seen in 15–89% — failure must not stop work-up.
- **VUDS**: SUI in up to 50%; true SUI vs paradoxic dribbling from sac; biphasic pressure profile.
- Imaging: **MRI best** (T2 high, no voiding, maps extent; may miss small cancers, cost); double-balloon PPU (Trattner) rarely available; ultrasound anechoic; VCUG needs voiding/postvoid film, underestimates size. Bulking agent on MRI: multiple, upper urethra, no fluid level/ostium, lower T2 than urine.

> **KEY POINT**
> UD symptoms mimic common LUTS and do not track size — suspicion and examination make the diagnosis; imaging must confirm and **map**, especially extension towards the bladder neck (bladder/ureteric injury, postoperative incontinence).

## 7 · Differential diagnosis of periurethral masses

| Lesion | Clue | Treatment |
|---|---|---|
| UD | Mid/proximal; communicates | Excision, reconstruction |
| Leiomyoma | Firm, mobile, solid; oestrogen-dependent | Transvaginal enucleation |
| Skene cyst/abscess | Lateral to and distorts meatus; non-communicating | Aspiration, marsupialisation, excision |
| Gartner duct cyst | Anterolateral; may receive ectopic ureter — image upper tract | Observe, sclerotherapy, excision |

| Lesion | Clue | Treatment |
|---|---|---|
| Vaginal wall cyst | Müllerian commonest | Excise if symptomatic |
| Urethral prolapse | Red "doughnut"; prepubertal, postmenopausal | Oestrogen; excision |
| Caruncle | Red, †posterior meatal lip; postmenopausal | Oestrogen; excise if atypical |
| Bulking agent | Injection history | Excise if symptomatic |

Skene gland adenocarcinoma may raise PSA.

**KEY POINT**
Two questions sort most periurethral masses: does it communicate (only UD), and where is it relative to the meatus (Skene, prolapse, caruncle at the meatus); firm mobile solid = leiomyoma.

## 8 · Classification and SUI

| System | Basis | Use |
|---|---|---|
| **L/N/S/C3** (Leach) | Location, Number, Size, Configuration, Communication, Continence | Standard; not validated |
| Ginsburg–Genadry | Location | Distal → marsupialisation; proximal → excision |
| Leng–McGuire | Periurethral fascia present? | Absent = **pseudodiverticulum** → flap/graft |

- De novo significant SUI **7–16%**; risk: ≥30 mm, proximal, circumferential, short urethra. Most settles within 12 months.
- ~50% of preoperative SUI improves after excision → concomitant surgery only for **bothersome** SUI: **autologous pubovaginal sling, never synthetic**.

## 9 · Surgical repair of UD

- Indications: symptoms, stones, suspected cancer. Asymptomatic/unfit: suppressive antibiotics + postvoid stripping.
- **Spence-Duckett marsupialisation**: distal-third only, not sexually active (SUI, vaginal voiding, dyspareunia). Unroofing/fulguration/oxidised cellulose abandoned; lap/robotic or suprameatal for anterior/proximal UD reported.
- Principles (Box 122.2): vascularised flap, separate fascia, excise ostium and entire sac, watertight, multilayer non-overlapping, close dead space, **preserve or create continence**.

**ALGORITHM**
Lithotomy, Foley; **SPC at the start** (never end — crosses repair) → **inverted-U incision** → raise vaginal flap off fascia → incise fascia **transversely**, raise flaps → excise entire sac to ostium incl. inflamed urethral edge → watertight urethral closure over catheter (as small as 12 Fr) → fascia **perpendicular**, obliterating dead space → ± **Martius** (poor tissue, attenuated fascia) → advance vaginal flap (3 layers; 4 with Martius) → antibiotic pack

- Pseudodiverticulum: Martius or fascial graft; circumferential may need segmental urethral resection.
- Post-op: antibiotics 24 h, antispasmodics; **pericatheter VCUG at 14–21 days**; leak → repeat weekly.

## 10 · Complications

| Complication | Incidence | Notes |
|---|---|---|
| Recurrent UD | **10–22%** (pooled ~10%) | Incomplete excision, dead space; circumferential, complex, proximal, multiple |
| Incontinence | 1.7–16% | Conservative 12 months, then autologous sling |
| Urethrovaginal fistula | 0.9–8.3% | Distal: may need no repair; proximal: repair ± Martius |
| Stricture | 0–5% | — |
| Recurrent UTI | up to 31% | Investigate for recurrence |
| Others | — | Distal urethral necrosis, bladder/ureteric injury, dyspareunia |

Redo excision: less symptom relief, more fistula. Persistent symptoms → image for recurrence or stricture.

> **KEY POINT**
> Complete sac removal, watertight closure, off-axis fascia and obliterated dead space all prevent the three failures: recurrence, fistula, incontinence.

| Feature | Bladder diverticulum | Female UD |
|---|---|---|
| Origin | Herniation through detrusor | Infected periurethral gland ruptured into lumen |
| Cause to treat | BOO / neurogenic | Infection; coexisting SUI |
| Commonest cancer | Urothelial (then SCC) | Adenocarcinoma |
| Key study | Cystoscopy + cytology; VCUG | MRI |
| Operate for | Complications, not size | Symptoms or complications |
| Operation | Diverticulectomy + outlet treatment | Transvaginal excision, layered reconstruction |

# 77 Sexual Function and Dysfunction in the Female

- P3 Female sexual dysfunction — *definition, prevalence, response models, DSM-5 vs ICSM/ISSWSH, causes, PLISSIT/FSFI, treatment overview by domain*

## 1 · Functional anatomy

**Distal vagina, clitoris and urethra** share blood supply and nerves and respond as one unit — the **clitoral complex**; urethral and anterior vaginal wall surgery therefore affects sexual function.

- **Clitoris**: penile homologue, **separate from urethra**. Two corpora (incomplete pectiniform septum) → crura on ischiopubic rami. Glans very sensitive but **not erectile**. Dorsal nerves **superficial on the dorsum**, outside tunica, divided by the deep suspensory ligament — at risk in hood/clitoral surgery.
- **Erectile**: clitoral body, crura, "vestibular" bulbs. **Non-erectile vascular** (↑ flow, no engorgement): labia minora, glans, periurethral tissue, vaginal walls. Urethra surrounded by erectile tissue except vaginally.
- **Skene** = prostate homologue (PSA-positive); **Bartholin** = bulbourethral homologue. MCQ G-spot has no agreed anatomical basis.
- Labium majus: anterior third ilioinguinal (L1); posterior two-thirds perineal (S3); lateral posterior cutaneous nerve of thigh (S2). Glans drains **directly to deep inguinal nodes**. Deep dorsal vein does not run with arteries.

## 2 · Physiology of the sexual response

> **ALGORITHM**
> Touch on clitoral complex → pudendal (somatic) afferents → **S2–S4** → parasympathetic efferents (nervi erigentes → cavernous nerves) → arteriolar dilatation → engorgement + lubrication

Reflex: **somatic afferents, visceral efferents**; voluntarily augmented or suppressed. **Orgasm**: cerebral, **sympathetic** (T12–L1, hypogastric) uterine contraction + pudendal pelvic striated muscle contraction. Resolution: sympathetic vasoconstriction.

> **KEY POINT**
> Pudendal injury removes the **trigger**; pelvic plexus/cavernous nerve injury removes the **response**.

> **ALGORITHM**
> nNOS → NO → guanylate cyclase → cGMP → ↓ $Ca^{2+}$ → smooth muscle relaxation → engorgement. PDE5 degrades cGMP; arginase consumes NOS substrate; VIP works with NO.

**Desire = excitation** (dopamine, noradrenaline, melanocortins, oxytocin, testosterone, oestrogen) vs **inhibition** (serotonin, opioids, prolactin, progesterone); licensed HSDD drugs act on this balance.

| | Oestrogen | Testosterone |
|---|---|---|
| Role | Maintains **substrate**: epithelium, collagen, lubrication, acidic pH, NOS/VIP | Drives **desire**; SHBG-bound is inactive |

| | Oestrogen | Testosterone |
|---|---|---|
| Deficiency | Thin, dry, fibrotic vagina → dyspareunia; risk ↑ at oestradiol **≤50 pg/mL** | ↓ Libido, arousal, sensation, orgasm |
| Treatment risk | — | Virilisation: early reversible (acne, hirsutism); late irreversible (voice, alopecia, clitoromegaly) |

| Model | Stages | Key idea |
|---|---|---|
| Masters & Johnson (1966), linear | Excitement → plateau → orgasm → resolution | **No desire phase** |
| Kaplan (1977), linear | Desire → arousal → orgasm | Adds desire |
| Whipple & Brash-McGreer, circular | Seduction → sensations → surrender → reflection | Reflection feeds next seduction |
| **Basson (2002), circular** | Neutrality → receptivity → arousal → **responsive desire** → satisfaction | Spontaneous desire **not needed**; intimacy the motive |

**KEY POINT**
Under Basson, lack of *spontaneous* desire (common after menopause) is not a dysfunction — the reason DSM-5 merged desire and arousal.

## 3 · Classification

DSM-5: **≥6 months + clinically significant distress**; low desire without distress is no disorder.

| Domain | DSM-IV-TR | DSM-5 |
|---|---|---|
| Desire + arousal | HSDD; FSAD; sexual aversion disorder | **FSIAD**: **≥3 of 6** — ↓ interest; ↓ fantasies; ↓ initiation/receptivity; ↓ pleasure, ↓ response to cues, ↓ sensation (in 75–100% of encounters). Aversion dropped |
| Orgasm | FOD after normal excitement | FOD: delay, infrequency, absence **or ↓ intensity** in **75–100%**; normal-excitement requirement removed |
| Pain | Dyspareunia; vaginismus | **GPPPD**, ≥1 of: penetration difficulty; pain; fear of pain; PFM tensing |

- **ICSM 2016** keeps HSDD and FSAD **separate**; adds **PGAD** and orgasmic illness syndrome (malaise after sex). **ISSWSH** splits FSAD into cognitive and **genital arousal disorder** (urological); not diagnosed if stimulation does not match her preferences.
- ~**40%** lifetime sexual problem; ~**8–12%** distressing disorder. **HSDD commonest**, FOD second.

## 4 · Causes (biopsychosocial)

- **Biological**: DM, hypertension, coronary disease, thyroid, hyperprolactinaemia, renal failure, SCI, MS, vulval dermatoses, incontinence, pregnancy, breastfeeding, menopause.
- **Psychological**: depression (bidirectional — screen every HSDD), anxiety, body image, abuse; restrictive upbringing; conflict, **partner's ED or PE**.
- **Drugs**: SSRIs, SNRIs, antipsychotics (↑ prolactin); β-blockers, ACE inhibitors, CCBs, thiazides, spironolactone; antiandrogens, GnRH agonists; opiates, alcohol. MCQ Combined hormonal contraceptives: ↓ ovarian testosterone + ↑ SHBG; oral oestrogens, tamoxifen, thyroxine also ↑ SHBG.

- **GSM**: dryness, burning, dyspareunia (often **vestibular**), urgency, dysuria, recurrent UTI; dryness → pain → low desire — **treat the tissue first**.
- **Cancer treatment**: castration, anti-oestrogens (†aromatase inhibitors → severe GSM), pelvic RT (stenosis), nerve injury.

## 5 · Evaluation

**Clinician must ask**; gender-neutral, do not assume penile–vaginal intercourse.

> ALGORITHM
> **PLISSIT**: **P**ermission → **L**imited **I**nformation (cycle, ageing, drugs) → **S**pecific **S**uggestions (lubricants, **vaginal oestrogen**, positions, vibrators) → **I**ntensive **T**herapy = **referral**

- **BSSC-W**: ACOG triage tool, **not validated**. **DSDS**: **validated**, 5 items; yes to 1–4 + no contributing cause = **generalised acquired HSDD**.
- **FSFI**: 19 items, 6 domains, most used. FSDS-R: **distress**. **PISQ-12/IR**: **only validated** tool for prolapse/incontinence. HSDD: PFSF postmenopause, SIDI premenopause.
- History: lifelong/acquired, generalised/situational, **order problems appeared** (primary vs secondary); drugs, surgery, LUTS, prolapse, **partner's** function.
- Examination (essential with pain): atrophy (pale, rugae lost, narrow introitus); **pH ≥4.5** without infection/semen = atrophy. **Q-tip test** of vestibule; **retract hood** for adhesions. Lichen sclerosus (white plaques → introital stenosis); lichen planus (**Wickham striae** pathognomonic → vaginal stenosis).
- PFM: **overactive** → dyspareunia; **underactive** → incontinence/prolapse. Fornix tenderness → endometriosis; ischial spine → pudendal neuropathy. Lukban–Whitmore tenderness 0 → 4 (prevents contraction). Cough stress test.
- Labs rarely: HbA1c, lipids if risk; prolactin/thyroid if signs. **No random testosterone** — no female range, unreliable assays, no level predicts HSDD. Photoplethysmography/Doppler research only.

## 6 · Hypoactive sexual desire disorder

- ISSWSH: ≥6 months loss of motivation/desire, **with distress**, not due to drug, illness, pain, abuse or partner. **Exclude relationship conflict first.** Coexisting arousal/orgasm disorder does not exclude HSDD.
- Desire and distress both fall with age → HSDD peaks in **midlife**; defined by distress, not hormone level (many HSDD women have normal oestradiol).

> ALGORITHM
> Screen (DSDS) → history + examination → **correct modifiable factors** (drugs, depression, GSM, pain, partner's ED) → education / sex or couples therapy → persistent **generalised acquired** HSDD → drug → review at 8 weeks. **Refer**: situational HSDD failing counselling, psychiatric illness, sexual trauma

| | Flibanserin | Bremelanotide |
|---|---|---|
| Mechanism | Central **5-HT1A agonist / 5-HT2A antagonist**: ↓ serotonin, ↑ dopamine/NA (removes brake) | Central **melanocortin agonist** → dopamine (presses accelerator) |
| Licence | FDA: only **premenopausal** acquired generalised HSDD | |

| | Flibanserin | Bremelanotide |
|---|---|---|
| Dose | **100 mg at bedtime** daily; ~half respond by 8 weeks | **1.75 mg subcutaneous** ≥45 min before sex; ≤1/24 h, **≤8/month** |
| Side effects | Dizziness, somnolence, nausea | Nausea, flushing, headache, **hyperpigmentation** |
| Cautions | **Alcohol** (hypotension, syncope); REMS; †strong CYP3A4 inhibitors | Uncontrolled hypertension/CVD |

- **Transdermal testosterone** (off-label): improves desire, arousal, orgasm, satisfaction in **postmenopausal HSDD**; oestrogen + androgen beats oestrogen alone; no CV signal, long-term safety unknown.
- †2019 Global Consensus: target **premenopausal physiological levels**; level at baseline and 3–6 weeks; **stop at 6 months** if no benefit; watch virilisation. Bupropion (off-label) alternative.

## 7 · Arousal disorder and PGAD

- **Subjective and genital arousal correlate poorly** — vasodilators ↑ flow without felt arousal (L-arginine + yohimbine). Partner-specific failure = relationship problem.
- Mindfulness-based sex-focused CBT; depression/anxiety predict poor response.
- **Transdermal testosterone** best evidence. PDE5i only with neurovascular impairment (type 1 DM, SCI, MS, SSRI-induced). **Bupropion 150 mg twice daily** for SSRI-induced dysfunction. Topical alprostadil ineffective; apomorphine small studies.

**PGAD (Leiblum–Nathan)**: genital arousal for hours–days, **not relieved by orgasm**, without desire, distressing. Tarlov cysts, pelvic varices, **stopping an SSRI**. CBT, duloxetine, pregabalin; surgery for lumbosacral annular tear.

## 8 · Female orgasmic disorder

- **Diagnosis of exclusion requiring distress**: exclude drugs, disease, stress, **inadequate stimulation** (incl. partner's ED/PE).
- **No orgasm at coitus without clitoral stimulation = normal variation.**
- Lifelong: inexperience, poor education; acquired: stimulation, context, physiology. Risks: anxiety, depression, sex guilt, conflict, abuse, DM, SCI, MS, cancer, SSRIs, antipsychotics, antiandrogens.
- **Education is the cornerstone**; directed masturbation, sensate focus, mindfulness. Arousal intact → more stimulation (vibrators; MCQ EROS clitoral vacuum = only FDA-cleared FSD device). Hormones if postmenopausal deficiency; surgery for clitoral phimosis/adhesions.

> **ALGORITHM**
> **SSRI-induced dysfunction**: present before treatment? → wait for tolerance (~4 months) → persists 3–6 months: lower dose or switch/add **bupropion** or mirtazapine; consider sildenafil

## 9 · Genito-pelvic pain / penetration disorder

- **Dyspareunia**: superficial or deep (endometriosis, prolapse, pelvic floor). **Vaginismus**: involuntary contraction preventing penetration.

- **Vulvodynia**: vulval (burning) pain **≥3 months**, no cause; localised/generalised, provoked/unprovoked.
- **Provoked vestibulodynia (PVD)** commonest (~**10–12%** of women; IBS, fibromyalgia). Primary: from first penetration (nulliparity, family history, enuresis, dysmenorrhoea). Secondary (~⅔): after pain-free sex; larger area, hood often most painful.
- Vestibular nerve proliferation + **hypertonic pelvic floor** + fear–avoidance.
- **Clitorodynia**: hood adhesions trap keratin → pseudocysts, pearls, pain — **examine glans routinely**.

> **ALGORITHM**
> Education + vulval care → lidocaine 5% or EMLA **10 min before intercourse** → **multimodal pelvic floor physical therapy** (myofascial release, trigger points, down-training, biofeedback, graded **dilators**, electrical stimulation) + psychological/couples therapy → vestibular oestrogen if hypo-oestrogenic → **vestibulectomy** for refractory localised PVD

- **Multimodal PT is standard of care** for PVD; EMG biofeedback best if pelvic floor overactive.
- Oestrogen deficiency ↑ vulval nerve density → **topical > systemic, vestibular > vaginal**.
- †Second line: amitriptyline, gabapentin, intralevator botulinum toxin.
- **Vestibulectomy** to within 1–2 mm of meatus; beats non-surgical therapy in localised neuroproliferative disease.

### Treatment of GSM

> **ALGORITHM**
> Non-hormonal lubricants/moisturisers + regular activity (NAMS first line) → **low-dose vaginal oestrogen** (†minimal absorption, no progestogen) or †vaginal DHEA (prasterone 6.5 mg nightly) or †ospemifene 60 mg orally (SERM, vaginal agonist; if vaginal products impossible; VTE risk) → systemic HT if vasomotor symptoms

> **KEY POINT**
> †After **breast cancer**: non-hormonal first; vaginal oestrogen only after oncology discussion (aromatase inhibitors).

## 10 · Pelvic surgery and sexual function

Three routes of harm: **nerves** (lateral vaginal wall, pelvic plexus), **clitoral blood supply** (distal urethra), **vaginal shape**. Counsel **before and after** — patients want it more than they ask.

- **Radical cystectomy**: loss of desire > orgasm > dyspareunia > lubrication; ~half can have intercourse. Preserve: (1) retrograde dissection saving NVBs on **lateral vaginal walls**; (2) divide urethra at **bladder neck** (clitoral supply); (3) keep vaginal dimensions. **Genital-sparing** (if oncologically safe): sex resumes **~6 vs 14 weeks**; diversion type irrelevant. A third of cancer patients want sex-therapy referral. **Never trade oncological control.** *→ Ch 101*
- **Hysterectomy**: benign — unchanged or better, route irrelevant; radical — lubrication/dyspareunia at 6 months, most active again by 12 months. **Rectal surgery**: dysfunction common after LAR, **nearly all** after APR; risks neoadjuvant chemo, low tumour, intersphincteric resection, Hartmann.

- **POP repair**: stage does not predict function (bother, body image do). Native tissue: most same/better; de novo dyspareunia possible. Dyspareunia: vaginal > abdominal; anterior mesh > **sacrocolpopexy** (apical gold standard). Biological grafts: more recurrence, abandoned. Mesh complications may appear years later.
- **Combined anterior + posterior repair ↑ dyspareunia** — stage in sexually active women. **Keep length, avoid excess skin excision, preserve introitus.** *→ Ch 119*
- **SUI surgery**: ~⅓ improve, most unchanged; **curing coital incontinence** predicts improvement. **TVT** may denervate vaginal wall; **TOT** spares dorsal clitoral nerve; PISQ-12 slightly favours TOT, FSFI no difference. Dyspareunia is a strong reason for sling removal.
- **Coital incontinence** (~1 in 10 incontinent): at penetration = stress; at orgasm = detrusor overactivity.
- FSD severity by LUTD: BPS > DO > urgency/mixed UI > SUI > dry OAB > voiding LUTS; HSDD pairs with SUI, orgasm problems with OAB. **BPS worst** (pain → fear → PFM overactivity → vulvodynia). Tolterodine improved all FSFI domains in OAB; **sacral neuromodulation** improves sexual function whatever the cause.
- **Genital cosmetic surgery**: poor evidence; screen for **body dysmorphic disorder**.

## 11 · Special populations

Ask orientation and gender identity; **trans male** = assigned female at birth. Fertility preservation before transition. Oestrogen: thromboembolism; testosterone: erythrocytosis (Hct **>50%**) — Hct and testosterone 3-monthly in year 1, then 1–2/year; spironolactone → potassium. SCI: impaired sensation, lubrication, orgasm, genital pain, spasticity. *→ Ch 126*

## 12 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| Flibanserin phase III | RCTs vs placebo, premenopausal | ↑ Satisfying events, ↑ desire, ↓ distress | First FDA-approved drug for HSDD |
| RECONNECT (bremelanotide) | 2 phase III RCTs | ↑ Desire, ↓ distress | On-demand option for premenopausal HSDD |
| Bupropion for SSRI-induced dysfunction (Cochrane) | Meta-analysis | Benefit at 150 mg twice daily | Add or switch to bupropion |

# ★ Putting it together: female pelvic decisions

## A · Incontinent woman

ALGORITHM

**1. Classify** (history, diary): SUI, UUI, MUI, overflow, functional. Stress test, mobility, POP, PVR, urinalysis (→ *Ch 105*).

**2. Urodynamics** not needed for uncomplicated demonstrable SUI (VALUE); use if mixed/unclear, voiding dysfunction, prior surgery, neurological, high-grade POP.

**3. Conservative first:** weight, fluids/caffeine, supervised PFMT, bladder training, pessary, postmenopausal vaginal oestrogen (→ *Ch 114*).

**4. SUI surgery** if failed (→ *Ch 116, 118*); UUI → drugs, botulinum toxin, neuromodulation.

## B · Choosing an SUI operation

| Operation | Best for | Drawbacks |
|---|---|---|
| Retropubic MUS | Standard; better for ISD | Perforation (cystoscopy mandatory), voiding dysfunction, rare vascular/bowel injury |
| Transobturator MUS | Avoiding retropubic space (prior surgery, obesity) | Groin/thigh pain; less effective in severe SUI |
| Autologous pubovaginal sling | Recurrent SUI, post-mesh, with urethral repair, mesh declined | Larger; more voiding dysfunction |
| Burch | Mesh declined; concurrent abdominal surgery | Posterior prolapse, de novo urgency |
| Bulking agent | Frail, minimal surgery | Lower, short-lived cure |

## C · Prolapse: which repair?

ALGORITHM

**POP-Q**: leading edge stages; II = within 1 cm of hymen (→ *Ch 105*).

Asymptomatic → observe; symptomatic → pessary or surgery.

**Support the apex**: uterosacral/sacrospinous suspension or sacrocolpopexy (most durable) (→ *Ch 117*).

**Frail, not sexually active** → colpocleisis.

**Occult SUI:** discuss concomitant sling (OPUS: fewer leaks, more complications).

## D · Vesicovaginal fistula

KEY POINT

Double-dye, upper tract imaging, biopsy after radiation/cancer. Post-hysterectomy: supratrigonal, mostly vaginal (Latzko or layered); abdominal if ureter involved, large/high, or failed vaginal repair. Tension-free watertight layers, non-overlapping lines, vascularised flap (Martius vaginally, omentum abdominally) for poor/irradiated tissue, continuous drainage, then cystogram (→ *Ch 121*).

## E · Mesh complications

**ALGORITHM**

1. Speculum + cystoscopy if suspected; may appear years later.
2. Small exposure → oestrogen ± trimming; larger/symptomatic → excise, close.
3. Mesh in bladder/urethra → remove, repair; no new mesh then.
4. Remove enough to solve the problem; never asymptomatic mesh (→ *Ch 119*).